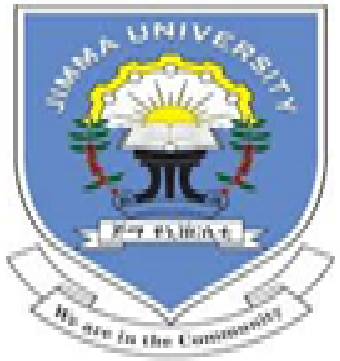


Chapter- Six

Model Evaluation

Compiled By: Alemisa E

Outline

- Model Evaluation
- Performance Evaluation methods
- Data processing
- Data cleaning and transforming
- Feature selection and visualization
- Model selection and tuning
- Methods of dimensional reduction
  - Principal component analysis (PCA)
  - Singular value decomposition (SVD)
  - T-distributed Stochastic Neighbor Embedding (t-SNE)

- Optimize the performance of the model.
- Control model complexity
- Cross-Validation and Re-sampling methods
  - K-Fold Cross-Validation
  - Bootstrapping
- Over-fitting and Under-fitting
- Gradient descent (batch, stochastic)
  - Bias, variance
- Tool kit

Model Evaluation

- **Model Evaluation** in ML refers to the process of assessing the performance of a machine learning model.
- It helps determine how well the model generalizes to ***new, unseen data*** and whether it is suitable for the specific task it is intended to solve.
- By ***assessing models***, we can choose the best one out of several options and learn more about their limits.
- The usage of data from ***testing*** and ***training*** is often involved in the evaluation process.
- A part of the available data is used during training to instruct the ***model*** in pattern ***recognition*** and ***prediction***.
- However, a different set of data that wasn't utilized during ***training*** must be used to ***test*** the model to assess its performance.

Key Objective of Model Evaluation

1.Measure Model Performance:

- *Quantify how well the model **makes predictions**.*

2.Detect Overfitting or Underfitting:

- Evaluate if the model is overly complex or too simple.

3.Compare Models:

- Assess multiple models to select the best-performing one.

4.Guide Improvements:

- Provide insights for tuning the model or optimizing the learning process.

Performance Evaluation Methods

- ***Model evaluation*** is the process that uses some metrics which help us to analyze the performance of the model.
  - ❖ *To evaluate the performance or quality of the model, different metrics are used, and these metrics are known as performance metrics or evaluation metrics.*
- These ***performance metrics*** help us understand how well our model has performed for the given data.
- Each **ML model** aims to generalize well on ***unseen/new data,*** and performance metrics help determine how well the model generalizes on the new dataset.
- ***Not all metrics*** can be used for all types of problems; hence, it is important to know and understand which metrics should be used.
- In this way, we can improve the model's performance by tuning the **hyper-parameters.**

Common Evaluation Metrics

- **For Classification Tasks:**

❑ ***Confusion Matrix***

  - *Accuracy*
  - *Precision*
  - *Recall (Sensitivity)*
  - *F1 Score*
  - *False Positive Rate (FPR)*
  - *ROC-AUC*

- **For Regression Tasks:**
  - Mean Squared Error (MSE)
  - Mean Absolute Error (MAE)
  - Root Mean Squared Error (RMSE)
  - R^2 (Coefficient of Determination)

- **For Clustering and Unsupervised Learning:**
  - Silhouette Score
  - Inertia:
  - Adjusted Rand Index

i. Evaluation Metrics for Classification Task

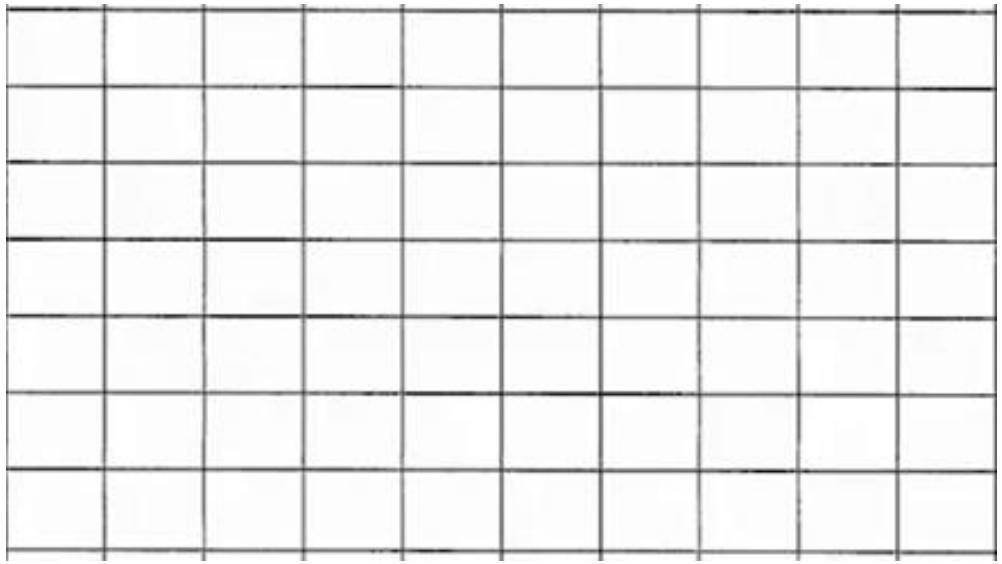
- In a *classification problem*, the category or classes of data are identified based on training data.
- It predicts class labels as the output, such as ***Yes or No, 0 or 1, Spam or Not Spam***, etc.
- Different metrics used for classification performance:
  - ***Confusion Matrix***
    - ***Accuracy***
    - ***Precision***
    - ***Recall***
    - ***F-Score***
    - ***False Positive Rate (FPR):***
    - ***AUC(Area Under the Curve)-ROC***

Key Metrics Derived from Confusion Matrix

Confusion Matrix

- A **confusion matrix** is a tabular representation used to evaluate the performance of a ***classification model.***
- A confusion matrix is ***n by n matrix , where n is represent the number of class in the variable.***

- ***n by n matrix***



Confusion Matrix

- Example in titanic dataset dependent classis only two variable.

Passenger ID	Survived	Pclass
1	0	3
2	1	1
3	1	3
4	1	1
5	0	3

- ***Here, the value of n= 2***

- *1=survive*
- *0=not survive*

Structure of the Confusion Matrix

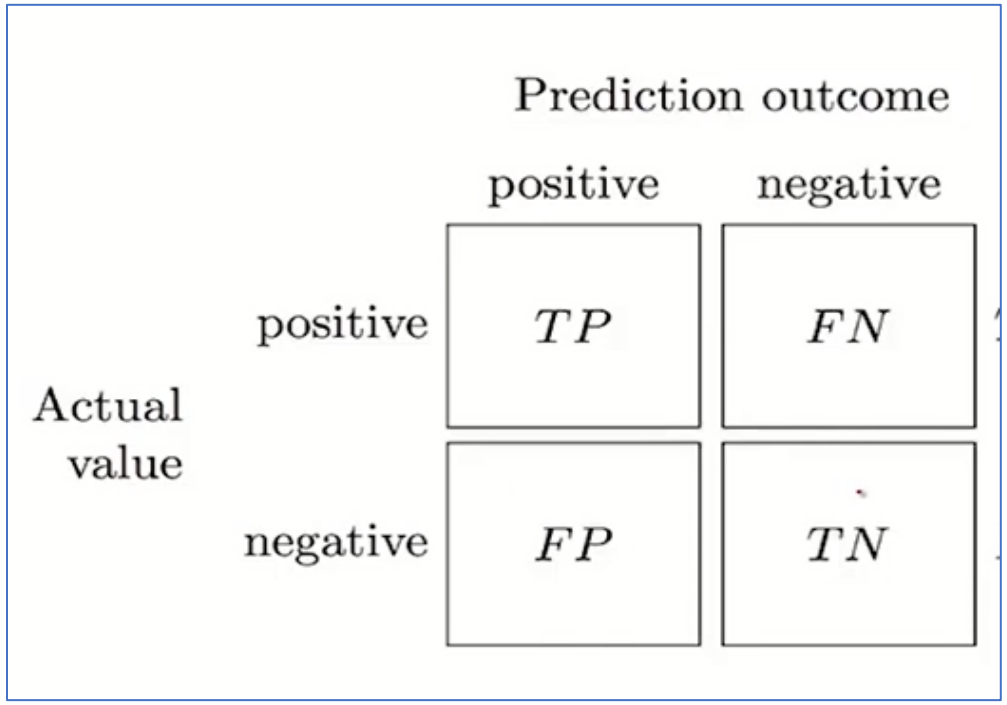
❑For a binary classification problem, the confusion matrix has four components:

- It compares the ***actual values*** with the ***predicted outcome***
- A confusion matrix is a table of at least size ***n by n*.**
- It can be understood well through a ***2×2 matrix*.**

❑This matrix consists of 4 main elements that show different metrics to count a number of correct and incorrect predictions.

❑ Each element has two words either as follows: True or False and Positive or Negative

Structure of the Confusion Matrix



$TP + FN$ Total Actual positive

$FP + TN$ Total Actual negative

- Where:
  - ***Row*** represents the ***actual value***,
  - ***Column*** represents ***the predicted values.***

Actual value	Predict outcome	
1	1	TP
1	0	FN
0	1	FP
0	0	TN

Confusion Matrix...

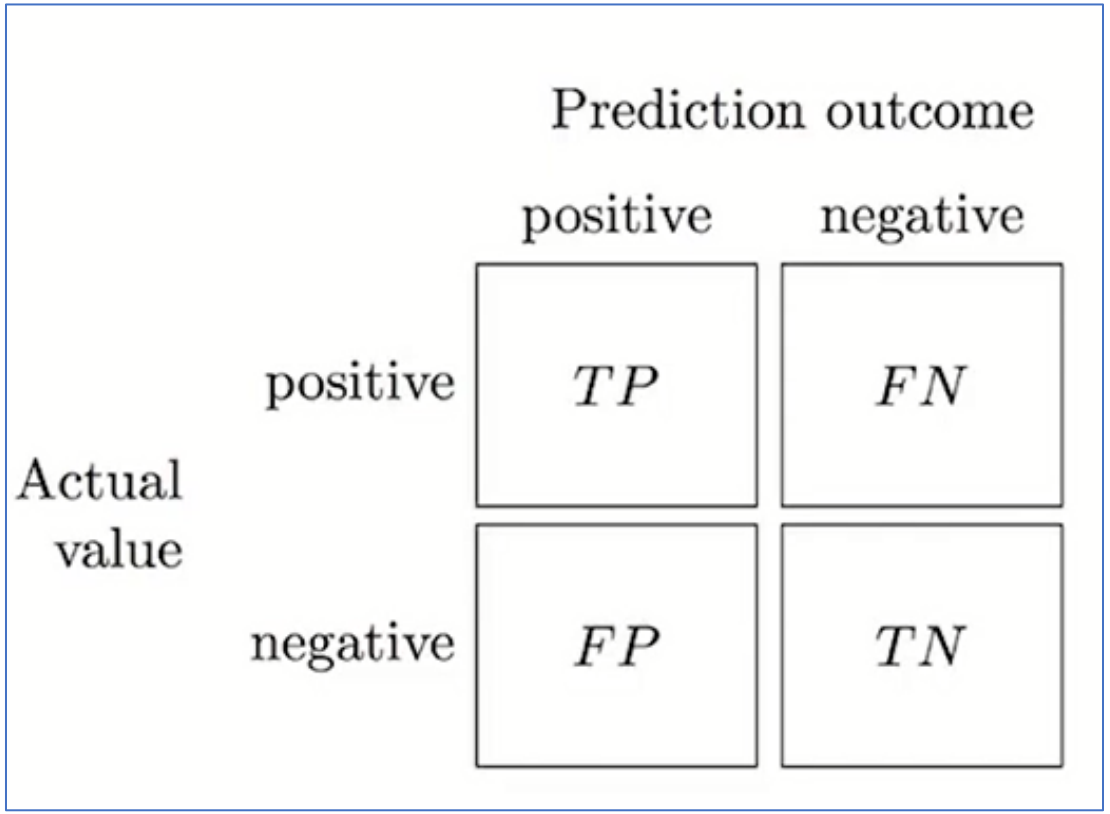
- If the predicted and truth labels match, then the prediction is said to be **correct,**
- **if the prediction is** mismatched, then the prediction is said to be ***incorrect.***
- Further, ***positive*** and **negative** represent the predicted labels in the matrix.

- Some terminologies in the matrix are as follows:

- ***True Positives:*** It is also known as **TP.**
  - ✓ It is the output in which the actual and the predicted values are **True**.
- **False Negatives:** It is also known as FN.
  - ✓It is the output in which the actual value is **True** but the predicted value is False.
- **False Positives:** It is also known as **FP.**
  - ✓It is the output in which the actual value is **False** but the predicted value is **True**.
- ***True Negatives:*** It is also known as **TN**.
  - ✓It is the output in which the actual and the predicted values are **False.**

Example 1: titanic dataset

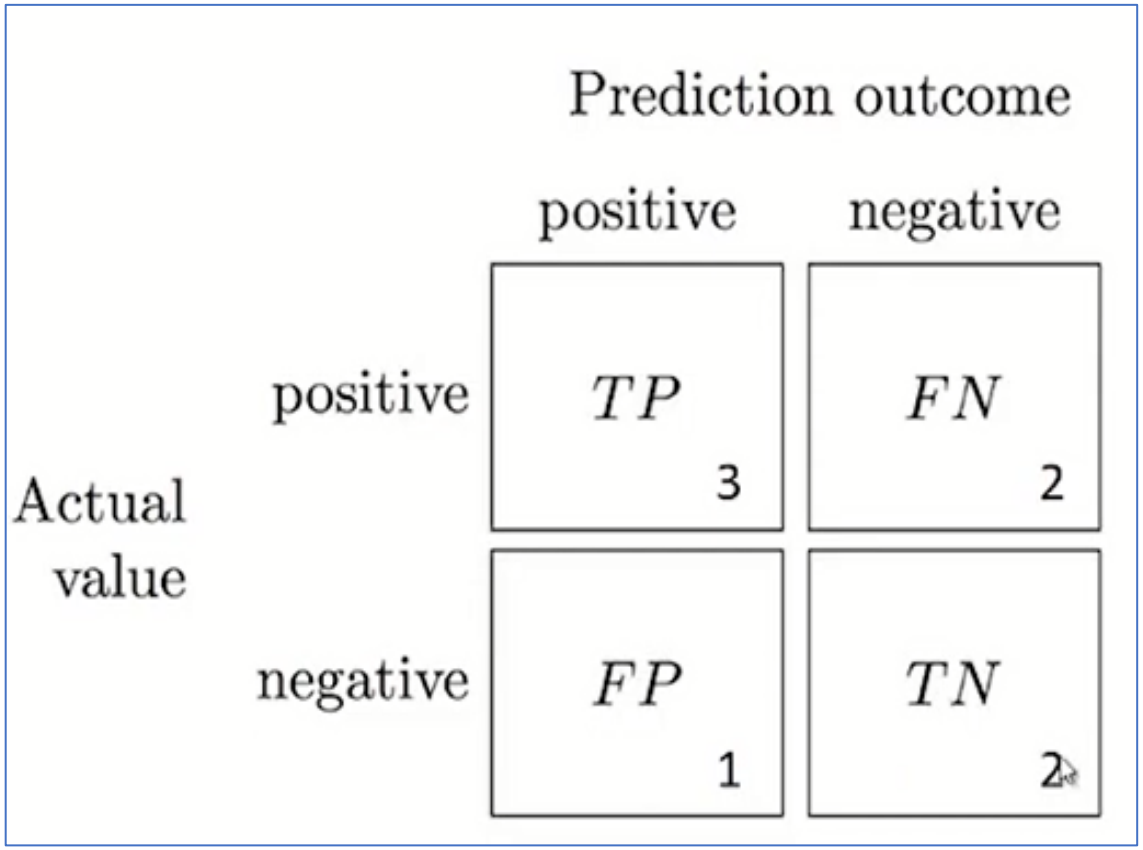
ID	Actual Survived	Predicted Survived?	Notation?
ID1	1	0	
ID2	1	1	
ID3	1	0	
ID4	0	0	
ID5	1	1	
ID6	1	1	
ID7	0	1	
ID8	0	0	



Cont'

ID	Actual Survived	Predicted Survived?	Notation?
ID1	1	0	**FN**
ID2	1	1	**TP**
ID3	1	0	**FN**
ID4	0	0	**TN**
ID5	1	1	**TP**
ID6	1	1	**TP**
ID7	0	1	**FP**
ID8	0	0	**TN**

- Next: Count the number of TP, FN, FPand TN



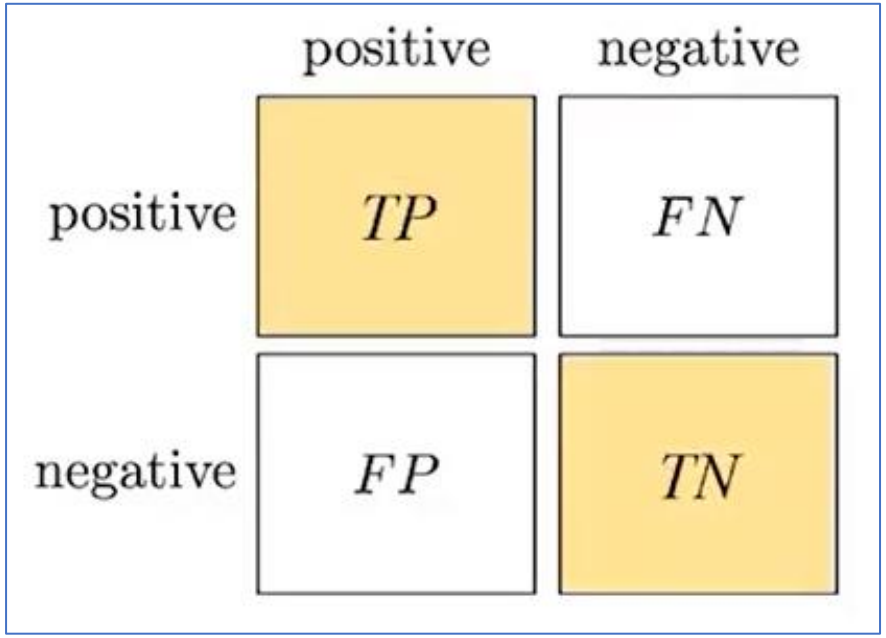
Accuracy

- **Accuracy** : It is the ratio of ***correct predicted values over the total predicted values.***
- It can be formulated as:

$$Accuracy = \frac{Number\ of\ Correct\ Predictions}{Total\ number\ of\ predictions}$$

- The formula is given by

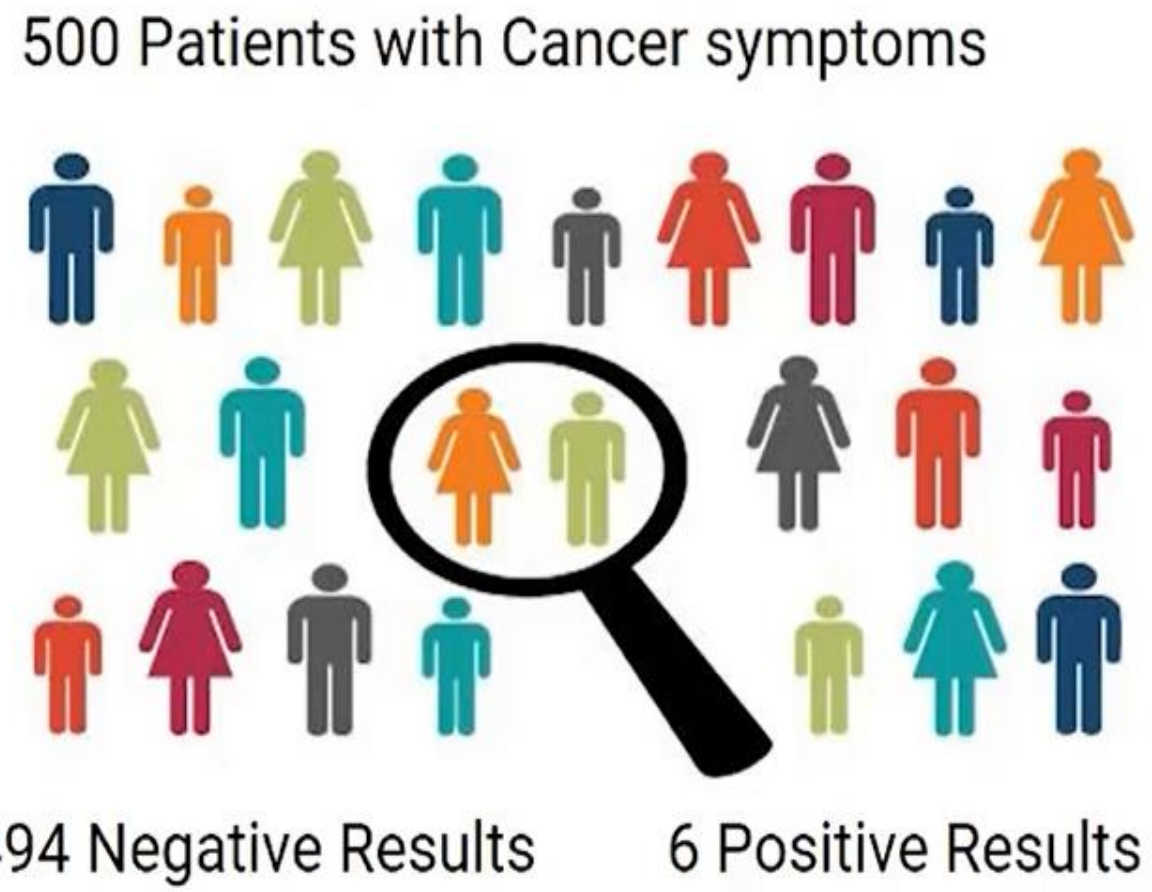
$$\boldsymbol{Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)}}$$



Example: **Accuracy**

We train a model to detect cancer

Actual value \ Prediction outcome	positive	negative
positive	4	2
negative	8	486



Example: calculate **Accuracy of our model**

We train a model to detect cancer

Actual value \ Prediction outcome	positive	negative
positive	4	2
negative	8	486

$$\text{Accuracy} = \frac{TP + TN}{TP + FN + FP + TN}$$

$$\boldsymbol{Accuracy = \frac{(4 + 486)}{(4 + 2 + 8 + 486)}}$$

$$\boldsymbol{Accuracy = \frac{490}{500}} = 0.98 = 98\%$$

Accuracy

- However, Accuracy has a drawback.
  - *It cannot perform well on an* **imbalanced dataset.**
  - *Suppose a model classifies that the majority of the data belongs to the major class label.*
  - *It yields higher accuracy. But in general, the model cannot classify on minor class labels and has poor performance.*

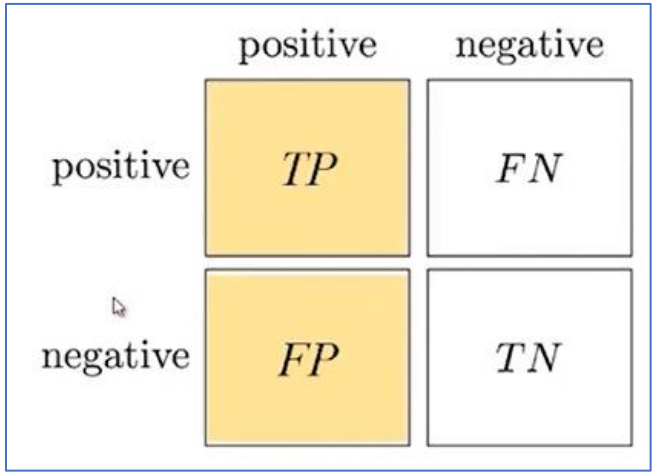
Precision

❑***Precision*** : is the *ratio of correctly classified positive predictions (True Positive) to the total number of classified positive samples* (either correctly or incorrectly).

❑i.e. Out of all the positive predictions, how many are actually positive.

$$\textbf{Precision} = \frac{\textbf{Predictions actually positive}}{\textbf{Total predicted positive}}$$

$$\textbf{Precision} = \frac{TP}{(TP + FP)}$$



❑The *drawback of Precision is* that it does not consider the ***True Negatives and False Negatives.***

Example: Evaluation Matrix -**Precision**

We train a model to detect cancer

		Prediction outcome: positive	Prediction outcome: negative
Actual value	positive	4	2
	negative	8	486



Example: Evaluation Matrix -**Precision**

We train a model to detect cancer

- Here, Accuracy is not good model

98% Accuracy for model

$$\text{precision} = \frac{TP}{TP + FP}$$

$$\boldsymbol{Precision} = \frac{\mathbf{4}}{\mathbf{(4+8)}} = 0.33 = 33\%$$

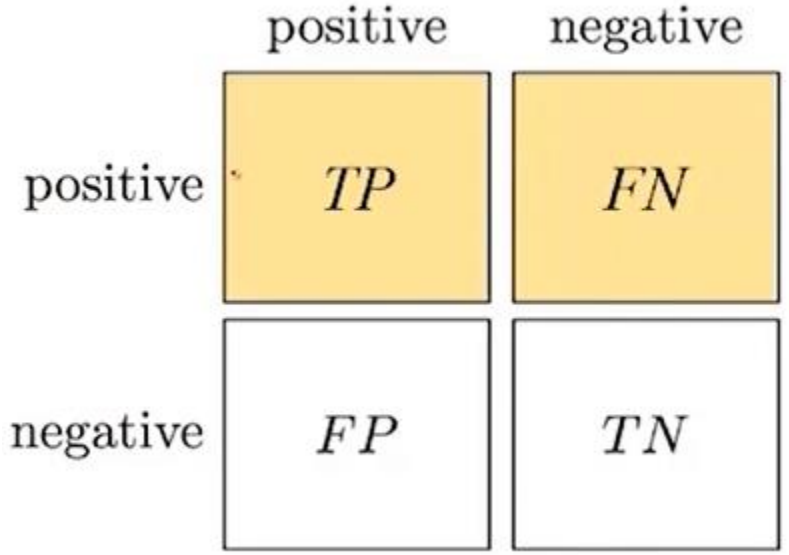
		Prediction outcome: positive	Prediction outcome: negative
Actual value: positive		4	2
Actual value: negative		8	486

Recall or Sensitivity

- **Recall** is out of all actual positive, how many are predicted positive.
- The ***recall measures the model's ability to detect positive samples***.

$$recall = \frac{Predictions\ Actually\ Positive}{Total\ Actual\ Positive}$$

$$\textbf{Recall} = \frac{\textbf{True Positive}}{\textbf{True Positive + False Negative}}$$



$$\textbf{Recall} = \frac{TP}{(TP+FN)}$$

- The ***drawback of Recall*** is that often it leads to a higher false positive rate.

F1 score

- **F-score or F1 Score -***the harmonic mean of both precision and Recall, assigning equal weight to each of them.*
- The formula for calculating the ***F1 score*** is given below:

$$\textbf{F1-score} = 2.\frac{(Precision\ x\ Recall)}{(Precision + Recall)}$$

- It is seen that during the *precision-recall* trade-off if we increase the *precision*, *recall decreases* and vice versa.
- The goal of the ***F1 score*** is to combine **precision** and **recall**.

False Positive Rate (FPR):

- The proportion of actual negatives that are incorrectly predicted as positive.

$$\mathbf{FPR} = \frac{\mathbf{FP}}{(\boldsymbol{FP+TN})}$$

Example #2

❑Suppose a binary classification model predicts whether a person has a disease (Positive) or not (Negative). If the model is evaluated on a dataset of 100 samples:

	Predicted Positive	Predicted Negative
Actual Positive	40	10
Actual Negative	5	45

- **True Positive (TP)** = 40
- **False Positive (FP)** = 5
- **False Negative (FN)** = 10
- **True Negative (TN)** = 45

$$\textbf{Accuracy} = \frac{(TP+TN)}{(TP+TN+FP+FN)} = \frac{40+45}{40+10+5+45} = 0.85 \text{ or } 85\%$$

$$Precision = \frac{TP}{(TP+FP)} = \frac{40}{40+5} = 0.89 \text{ 0r } 89\%$$

$$\textbf{Recall} = \frac{TP}{(TP+FN)} = \frac{40}{40+10} = 0.80 \text{ or } 80\%$$

$$\textbf{F1–score} = 2.\frac{(Precision\ x\ Recall)}{(Precision\ x\ Recall)} = \frac{0.89\ X\ 0.80}{0.89+0.80} \approx 0.84$$

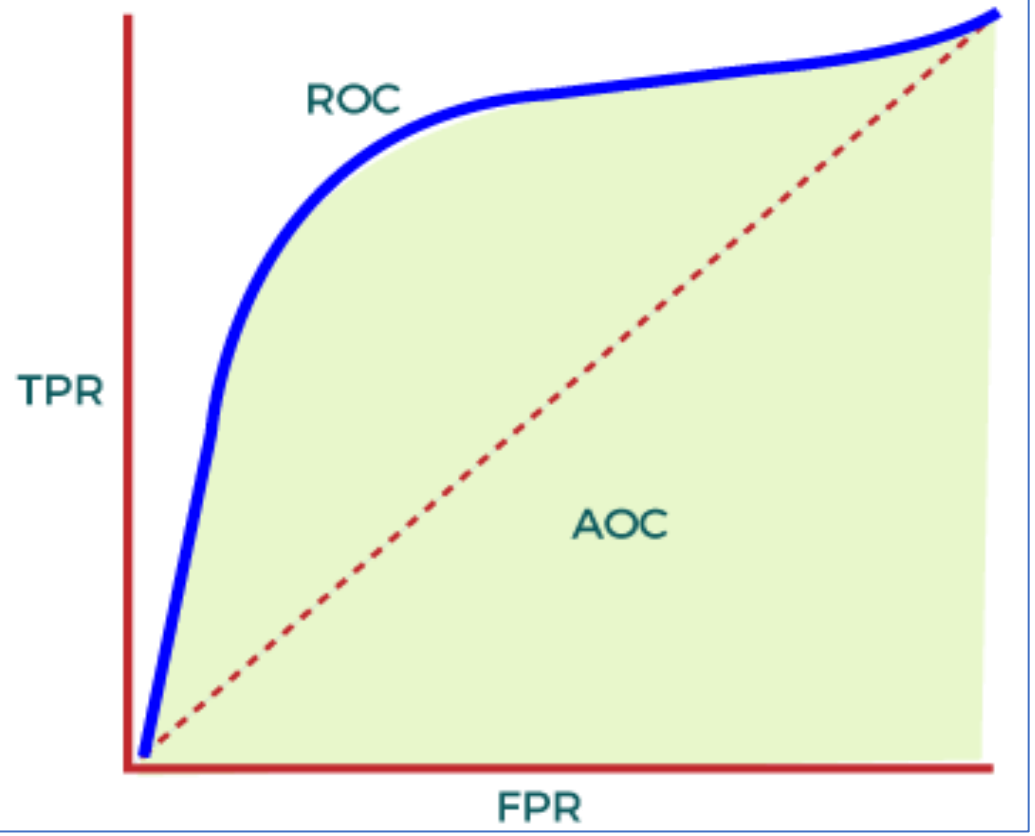
AUC-ROC Curve

- Sometimes we need to visualize the performance of the classification model on charts; then, we can use the ***AUC-ROC curve***.
- It is one of the popular and important metrics for evaluating the performance of the classification model.
- **AUC (Area Under Curve)**
  - is an evaluation metric that is used to analyze the classification model at different *threshold values.*
- **The Receiver Operating Characteristic(ROC) curve**
  - is a probabilistic curve used to highlight the model's performance.

AUC: Area Under the ROC curve

- The curve is plotted between two parameters, which are:
- **TPR:** It stands for True positive rate.
- It follows the formula of **Recall.**

$$TPR = \frac{TP}{TP + FN}$$



- **FPR:** It stands for False Positive rate.

$$\mathbf{FPR} = \frac{\mathbf{FP}}{(\boldsymbol{FP+TN})}$$

- This curve is useful as it helps us to determine the model's capacity to distinguish between different classes.

ii. Evaluation Metrics for Regression Task

- Regression tasks involve predicting **continuous values**, and their evaluation metrics measure how close the ***predicted values*** are to the ***actual values***.
- It is mostly used to find a relation between a dependent and an independent variable.
- For regression analysis, we are predicting a numerical value that may differ from the ***actual output***.
- So we consider the ***error calculation*** as it helps to summarize how close the **prediction** is to the ***actual value***.
- The metrics used for regression are different from the classification metrics.

ii. Common Evaluation Metrics for Regression Task

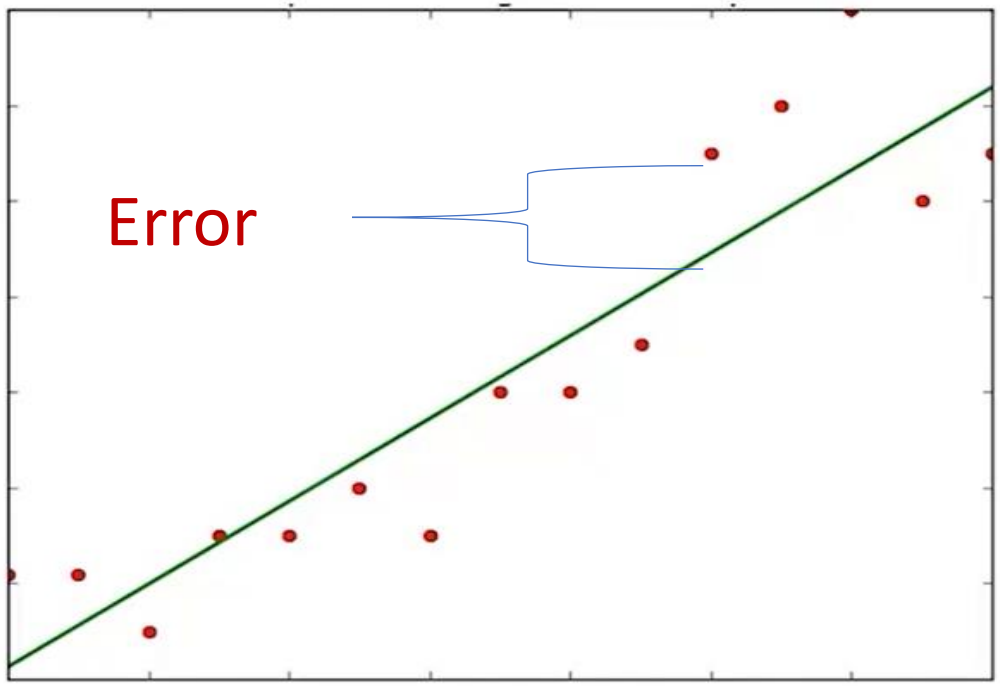
1. Mean Absolute Error,
2. Mean Squared Error
3. Root Mean Squared Error (RMSE)
4. R-squared (R^2 or Coefficient of Determination)
5. Adjusted R Squared
6. Mean Absolute Percentage Error (MAPE)
7. Explained Variance Score
8. Huber Loss
9. Mean Bias Deviation (MBD)

What is Error??

- **Error** determine how far the observe value from the predicted value.
- E.g. Look at simple linear regression

Actual Values	Predicted Values	Error
19	28	9
37	33	-4
25	20	-5
9	16	7
22	15	-7

e.g. for each at point



1. Mean Absolute Error (MAE)

- *MAE* is one of the simplest metrics, which measures the absolute difference between actual and predicted values,
- where ***absolute*** means taking a number as ***Positive***.
- The formula to calculate MAE:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

Where:

- y_i: Actual value.
- $\hat{y}_i$: Predicted value.
- n: Total number of observations.

- ***MAE*** is much more robust for the outliers.
- One of the limitations of MAE is that it is not differentiable
- However, to overcome this limitation, another metric can be used, which is *Mean Squared Error or MSE.*
- Use: Measures the average magnitude of errors without considering their direction.
- Range: [0, ∞] (lower is better).

2. Mean Squared Error (MSE)

- **MSE** measures the average of the Squared difference between predicted values and the actual value given by the model.
- Since in ***MSE***, errors are squared, therefore it only assumes *non-negative values*, and it is usually *positive and non-zero*.

- **Formula:**

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Where:

- y_i: Actual value.
- $\hat{y}_i$: Predicted value.
- n: Total number of observations.

- **Use:** Penalizes large errors more than MAE because of squaring. Sensitive to outliers.
- **Range:** [0, ∞] (lower is better).
- Moreover, due to squared differences, it penalizes small errors also, and hence it leads to over-estimation of how bad the model is.
- **MSE** is a much-preferred metric compared to other regression metrics as it is differentiable and hence optimized better.

3. Root Mean Squared Error (RMSE)

- RMSE: the square root of MSE, providing error in the same units as the target variable.
- **Formula**:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Where:

- y_i: Actual value.
- $\hat{y}_i$: Predicted value.
- n: Total number of observations.

❑ Use: Like MSE, but easier to interpret due to the same scale as actual values.
❑ Range: [0, ∞] (lower is better).

4. R-squared (R^2 or Coefficient of Determination)

- **R squared** is also known as Coefficient of Determination.
- Measures the proportion of variance in the target variable explained by the model.
- The **R-squared metric** enables us to compare our model with a constant baseline to determine the performance of the model.

- Formula:

$$R^2 = 1 - \frac{MSE(Model)}{MSE(Baseline)} = R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

- **Use:** Indicates how well the model explains the data.
- **Range:** [0, 1] (closer to 1 is better)

- $R^2 = 1$: Perfect model.
- $R^2 = 0$: Model is no better than the mean.
- Negative R^2: Model performs worse than the mean.

5. Adjusted R Squared

- **Adjusted R squared**, as the name suggests, is the improved version of R squared error.
- **R square** has a limitation of improvement of a score on increasing the terms
- To overcome the issue of R square, ***adjusted R squared*** is used, which will always show a lower value than R^2.
- It is because it adjusts the values of increasing predictors and only shows improvement if there is a real improvement.

- Formula:

$$R^2_{adj} = 1 - \left(\frac{(1 - R^2)(n - 1)}{n - p - 1} \right)$$

- Where ***n*** is the number of observations, and **p** is the number of predictors.
- Use: Useful when comparing models with different numbers of predictors.

6. Mean Absolute Percentage Error (MAPE)

- The average percentage error between predicted and actual values.
- Formula

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100$$

- Use: Provides an interpretable percentage error.
- Range: [0, ∞] (lower is better).

7. Explained Variance Score

- Measures the proportion of the variance in the target variable explained by the predictions.
- Formula:

$$\text{Explained Variance} = 1 - \frac{\text{Variance of residuals}}{\text{Variance of actual values}}$$

- **Range:** [0, 1] (closer to 1 is better).

8. Huber Loss

❑ A combination of MAE and MSE that is less sensitive to outliers than MSE.

❑**Formula:**

$$L_\delta(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2 & \text{if } |y - \hat{y}| \leq \delta \\ \delta|y - \hat{y}| - \frac{1}{2}\delta^2 & \text{otherwise} \end{cases}$$

❑**Where**

- *δ is a threshold.*
- *Use: Balances robustness to outliers with efficient convergence.*

9. Mean Bias Deviation (MBD)

- Measures the average bias in predictions.
- **Formula**:

$$MBD = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)$$

- **Use**: *Evaluate whether predictions are systematically overestimating or underestimating.*

Examples:
Solved regression evaluation metrics using sample data

❑**Sample Data**:

- Actual values (y): $[3, -0.5, 2, 7]$
- Predicted values ($\hat{y}$): $[2.5, 0.0, 2, 8]$

1. Mean Absolute Error (MAE)

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

$$MAE = \frac{|3 - 2.5| + |-0.5 - 0.0| + |2 - 2| + |7 - 8|}{4}$$

$$MAE = \frac{0.5 + 0.5 + 0 + 1}{4} = \frac{2}{4} = 0.5$$

Solution…

2. Mean Squared Error (MSE)

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$MSE = \frac{(3-2.5)^2 + (-0.5-0.0)^2 + (2-2)^2 + (7-8)^2}{4}$$

$$MSE = \frac{(0.5)^2 + (0.5)^2 + (0)^2 + (-1)^2}{4} = \frac{0.25+0.25+0+1}{4} = \frac{1.5}{4} = 0.375$$

3. Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{MSE}$$

$$RMSE = \sqrt{0.375} \approx 0.612$$

Solution…

4. R-squared (R^2)

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

- $\bar{y}$: Mean of actual values.

$$\bar{y} = \frac{3 + (-0.5) + 2 + 7}{4} = \frac{11.5}{4} = 2.875$$

- Compute numerator ($\sum(y_i - \hat{y}_i)^2$):

$$\sum(y_i - \hat{y}_i)^2 = (0.5)^2 + (0.5)^2 + (0)^2 + (-1)^2 = 0.25 + 0.25 + 0 + 1 = 1.5$$

Solution…

- Compute denominator $(\sum(y_i - \bar{y})^2)$:

$$\sum(y_i - \bar{y})^2 = (3 - 2.875)^2 + (-0.5 - 2.875)^2 + (2 - 2.875)^2 + (7 - 2.875)^2$$

$$= (0.125)^2 + (-3.375)^2 + (-0.875)^2 + (4.125)^2 = 0.015625 + 11.390625 + 0.765625 + 17.015625 = 29.1875$$

- Compute R^2:

$$R^2 = 1 - \frac{1.5}{29.1875} \approx 1 - 0.0514 = 0.9486$$

Solution…

5. Mean Absolute Percentage Error (MAPE)

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100$$

$$MAPE = \frac{1}{4}\left(\left|\frac{3-2.5}{3}\right| + \left|\frac{-0.5-0.0}{-0.5}\right| + \left|\frac{2-2}{2}\right| + \left|\frac{7-8}{7}\right|\right) \times 100$$

$$MAPE = \frac{1}{4}\left(0.1667 + 1.0 + 0.0 + 0.1429\right) \times 100$$

$$MAPE = \frac{1.3096}{4} \times 100 = 32.74\%$$

Results:

- MAE: 0.5
- MSE: 0.375
- RMSE: 0.612
- R^2: 0.9486
- MAPE: 32.74%

When to Use Each Metric

- ***MAE*:** When interpretability of errors is key, and outliers should not dominate.
- ***MSE/RMSE:*** When penalizing larger errors more heavily is desired.
- **R^2:** To understand how well the model explains variability.
- ***MAPE***: When relative error (percentage) matters.
- ***Huber Loss:*** For datasets with significant outliers.

Data Processing

❑ What is Data Processing?



❑***Data processing*** is a key component of ML,

❑***Is a process of preparing the raw data and making it suitable for a Machine learning model.***

❑When creating a machine learning project, we come across clean and formatted data.

❑While doing any operation with data, it is mandatory to clean it and put it in a formatted way. So for this, we use a ***data preprocessing task.***

Why do we need to Preprocess Data?



- Because, real-world data generally contains ***noises, missing values, Inconsistent,*** and maybe in an unusable format which cannot be directly used for ML models.

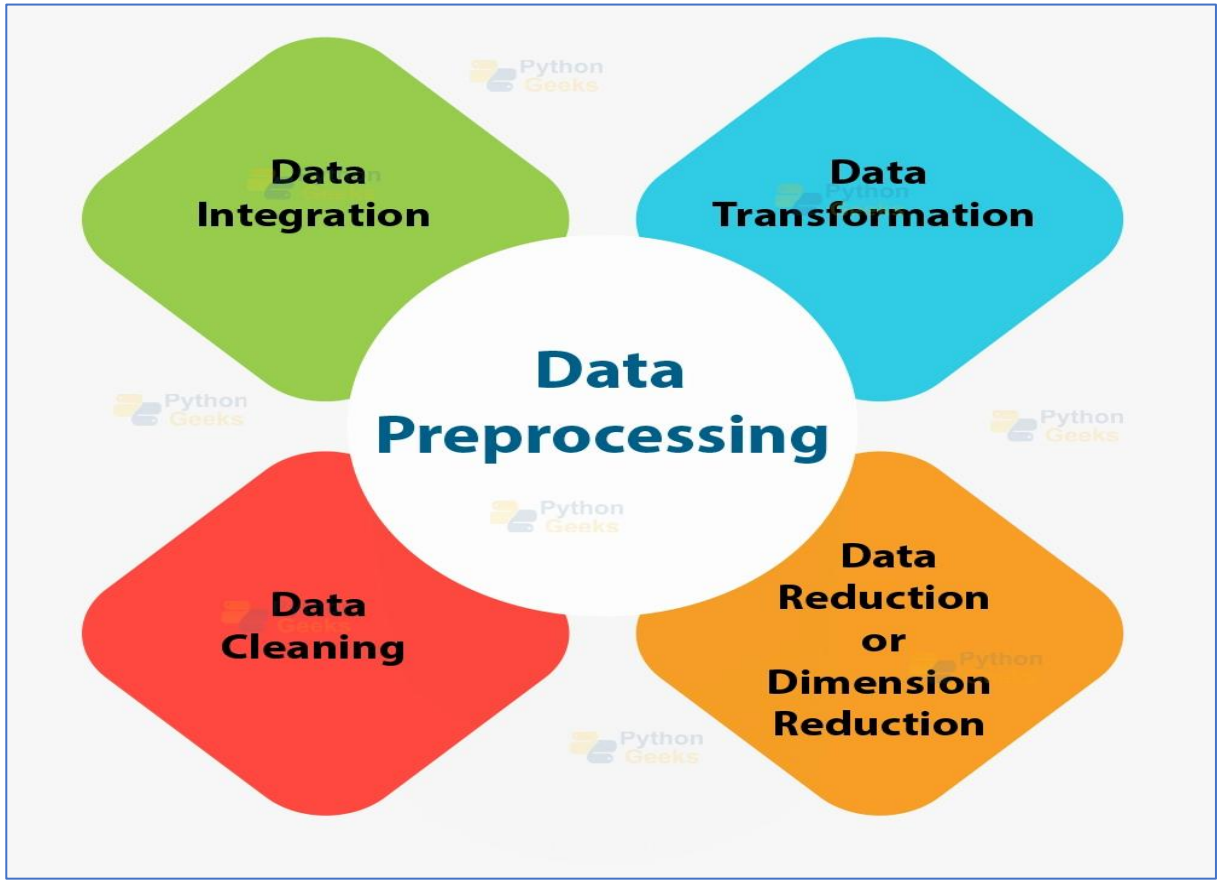
- ***Data preprocessing*** is a required task for cleaning the data and making it suitable for a ML model, which also increases the accuracy and efficiency of a ML model.

- Quality decisions must be based on quality data.
- **Data preprocessing is** important to get this quality data, without which it would just be a ***Garbage In , Garbage Out scenario.***

Methods of data processing

❑ To use the data for ***training a model,*** it must first be transformed and prepared, which is known as ***data processing.***

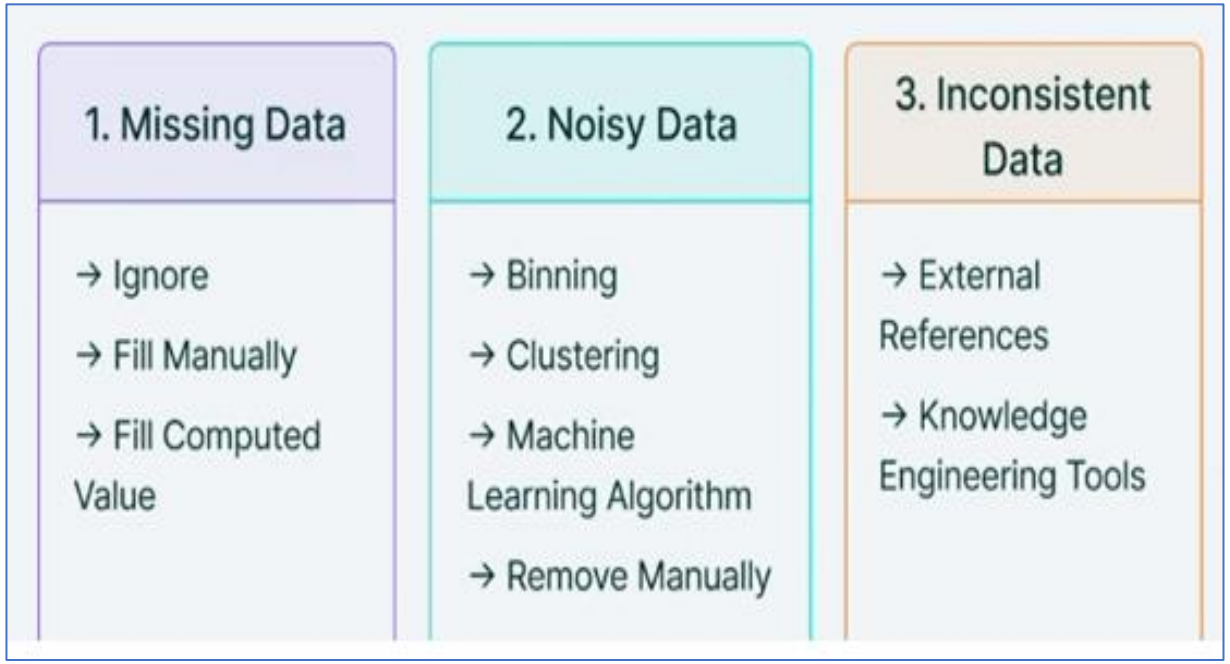
❑Here are methods of data processing:



Data Cleaning

- **Data cleaning** is the first phase, which identifies and fixes mistakes, discrepancies, dealing with outliers & and missing values in the dataset to improve data quality.

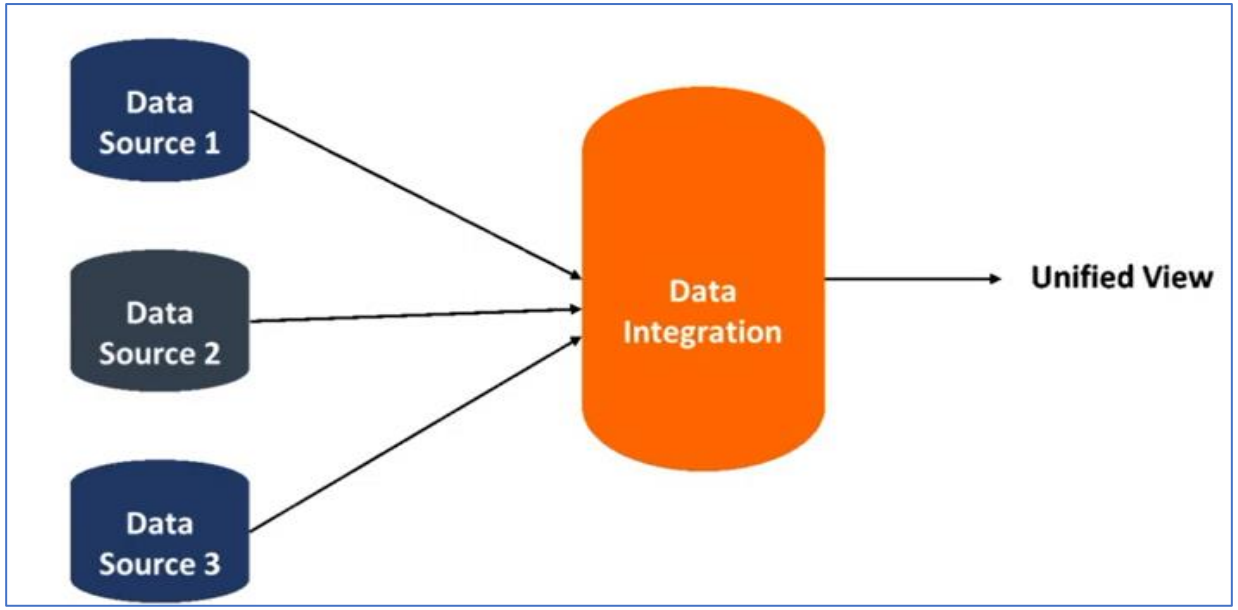
How to do that??



- ***Note**: The mean, median, or interpolation approaches can be used to impute missing data.*

Data Integration

❑When data is gathered from several sources, integration is necessary to combine and consolidate the ***datasets*** into a single, comprehensive dataset.



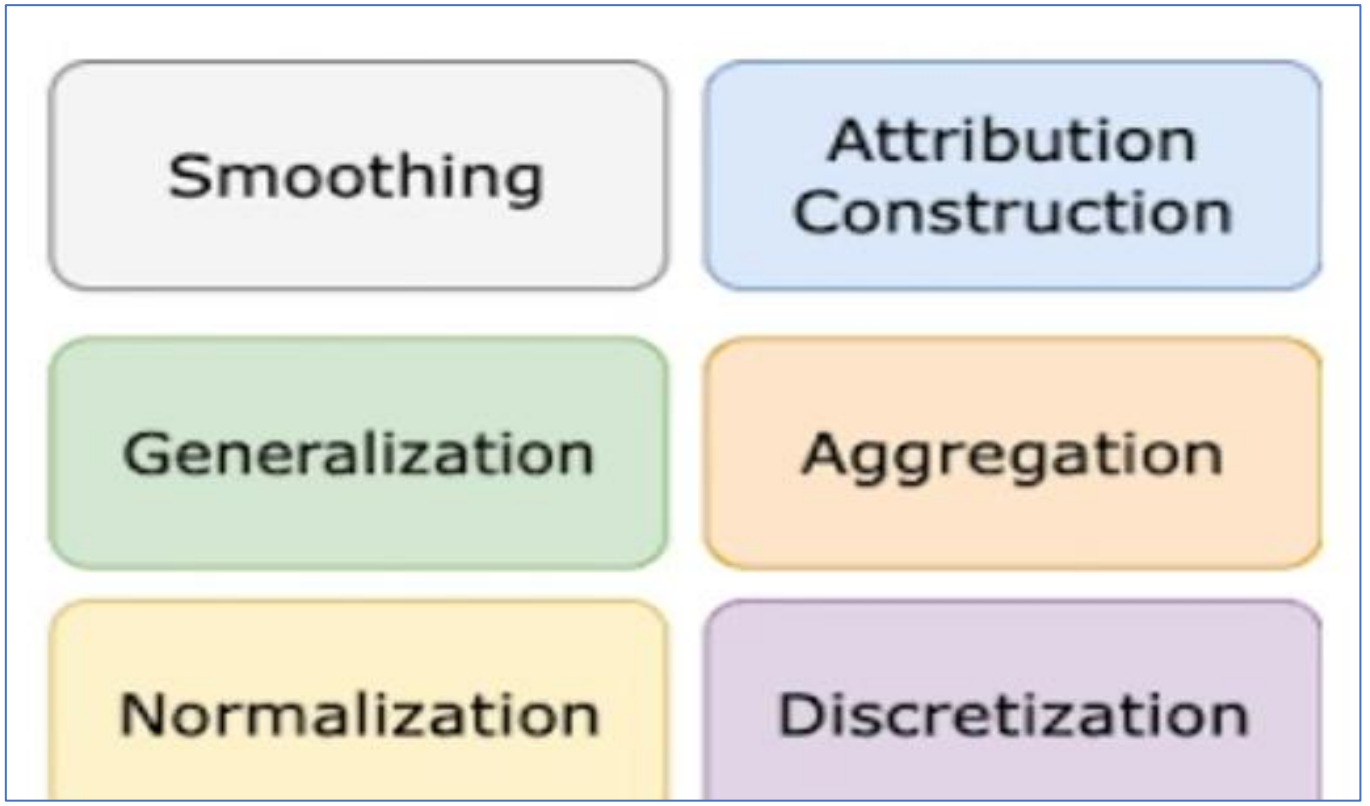
❑ *The process of merging data from several sources or formats into a single cohesive dataset is known as* ***data integration.***

❑ It entails addressing discrepancies, dealing with duplicate records, and merging data based on shared keys or identifiers.

Data Transformation

❑**Data transformation** entails changing or scaling the data to satisfy the presumptions or specifications of the selected machine learning algorithm.

❑Common methods :



Data Reduction or Dimension Reduction

- Refers to the process of reducing the number of input variables (features) in a dataset while retaining as much of the original information as possible.

There are two main types of **data reduction**:

1.Feature Selection: The process of selecting a subset of the most relevant features from the original set.

2.Feature Extraction: The process of transforming the data into a new set of features, typically with lower dimensions, while retaining important information.

Data Processing Cycle

- The continual process of transforming & analyzing data to train and enhance machine learning models.

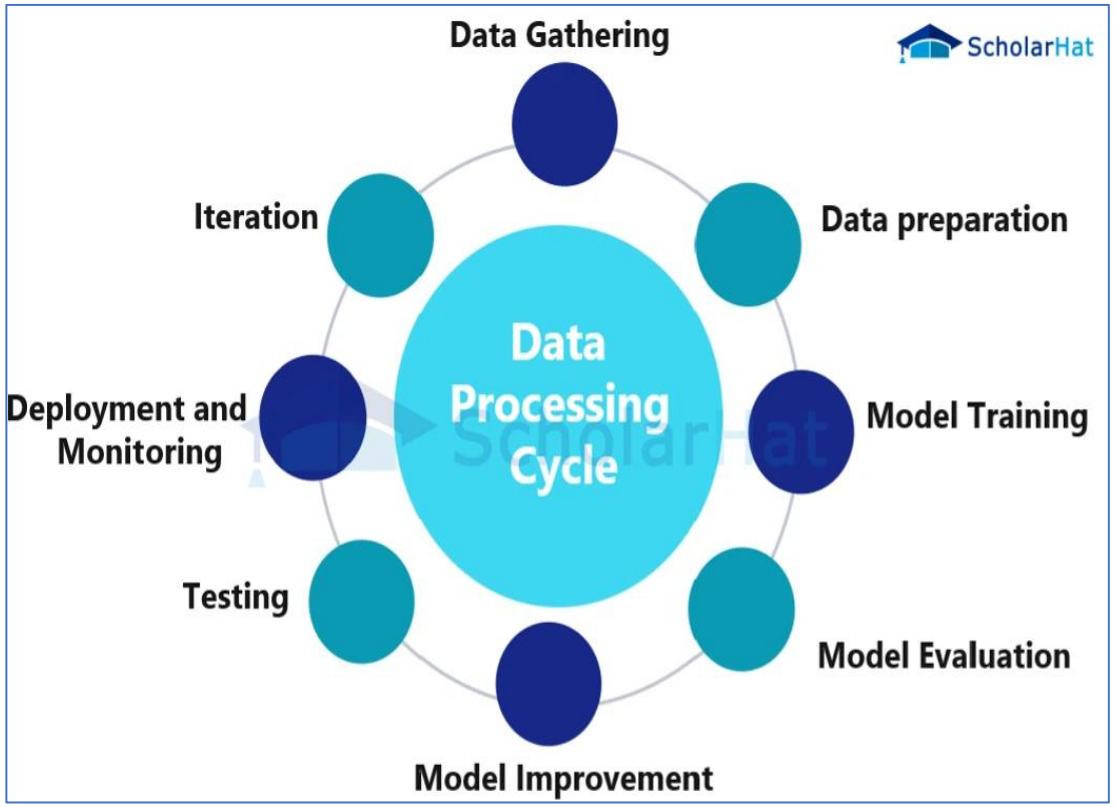


Fig: Data Processing Cycle

Data Processing Cycle

1.Data Gathering:

- Gathering pertinent data from numerous sources, including databases, files, and online platforms, is the initial stage.
- Depending on the particular issue, the data may include *characteristics, labels, or target variables.*

2. Data preparation:

- In this process, the data is cleaned to get rid of mistakes, discrepancies, and missing values.

Data Processing Cycle

3. Model Training:

- During this phase, a model based on ML is trained using the preprocessed data.
- The model attempts to identify patterns and connections between the features as well as the target variable by learning from the input data.

4. Model Evaluation:

- During this phase the effectiveness of the model is assessed using cross-validation methods or validation data.

5. Model Improvement:

- Changes are made to the model's operation based on the findings of the evaluation.

Data Processing Cycle

6. Testing:

- This step offers an unbiased evaluation of the model's capacity for generalization as well as how well it operates in previously untested situations.

7. Deployment and Monitoring:

- The model can be deployed for use in real-world applications if it operates well on the test data.

8. Iteration:

- Since data processing cycles are frequently iterative, the aforementioned stages are repeated as necessary.
- To continuously develop the model and react to shifting circumstances, new data can be gathered, & the preprocessing, training, evaluation, & improvement phases can be repeated.

How to get datasets for Machine Learning

❑ ML depends vigorously on datasets for preparing models and making precise predictions.

❑ What is a Dataset?

- ❖ **A dataset** is a collection of data in which data is arranged in some order.
- ❖ A ***dataset*** can contain any data from a series of an array to a database table.
- ❖ The table shows an example of the dataset:

Country	Age	Salary	Purchased
India	38	48000	No
France	43	45000	Yes
Germany	30	54000	No
France	48	65000	No
Germany	40		Yes
India	35	58000	Yes

- ❖ *Where each column corresponds to a **particular variable,** and each row corresponds to the **fields of the dataset.***
- ❖ *The most supported file type for a tabular dataset is **"Comma Separated File,"** or **CSV.***
- ❖ *But to store "tree-like data," we can use the JSON file more efficiently.*

Dataset vs Database

❑Dataset

- ❖A *dataset* is a collection of data that is usually stored in a **flat-file format, such as a CSV or Excel file.**
- ❖A dataset may contain data from a variety of *sources, including databases, websites, and manual entry.*

❑*Database*

- ❖Database is a collection of data that is organized and stored in a way that allows easy access and retrieval.
- ❖There are many different types of databases, including ***relational databases, object-oriented databases, and NoSQL databases.***

Types of data in datasets

- **Numerical data:**
  - Such as house price, temperature, etc.
- **Categorical data:**
  - Such as Yes/No, True/False, Blue/green, etc.
- ***Ordinal data:***
  - These data are similar to categorical data but can be measured on the basis of comparison.
- ***Note:*** *A real-world dataset is of huge size, which is difficult to manage and process at the initial level. Therefore, to practice machine learning algorithms, we can use any dummy dataset.*

Types of datasets

- A few normal sorts of datasets utilized in machine learning include:
- **Image Datasets:**
  - *Image datasets* contain an assortment of images and are normally utilized in computer vision tasks such as ***image classification, object detection, and image segmentation.***
  - **Examples:** ImageNet, CIFAR-10, MNIST
- **Text Datasets:**
  - Text datasets comprise textual information, like articles, books, or virtual entertainment posts.
  - These datasets are utilized in NLP techniques like ***sentiment analysis, text classification, and machine translation.***
  - **Examples:** Gutenberg Task dataset, IMDb film reviews dataset

Types of datasets

❑Time Series Datasets:

- ❖This type of dataset includes information focuses gathered after some time.
- ❖They are generally utilized in determining, abnormality location, and pattern examination.
- ❑ **Examples:** ***Securities exchange information, Climate information, Sensor readings.***

❑Tabular Datasets:

- ❑Tabular datasets are organized information coordinated in tables or calculation sheets.
- ❑They contain lines addressing examples or tests and segments addressing highlights or qualities.

Popular sources for ML datasets

1. Kaggle Datasets

- ***Kaggle*** is one of the best sources for providing datasets for Data Scientists and Machine Learners.
- The link is https://www.kaggle.com/datasets.

2. UCI Machine Learning Repository

- UCI stands for the University of California Irvine ML repository, and it is a very useful resource for getting open-source and free datasets for ML.
- It contains a huge collection of datasets sorted by ML tasks such as ***regression, classification, and clustering.***
- The link is https://archive.ics.uci.edu/ml/index.php.

Popular sources for ML datasets

3. Datasets via AWS (Amazon Web Services)

- We can search, download, access, and share the datasets that are publicly available via AWS resources.
- These datasets can be accessed through AWS resources but provided and maintained by different *government organizations, researchers, businesses, or individuals.*
- Anyone can add any dataset or example to the **Registry of Open Data on AWS.**
- The link is https://registry.opendata.aws/.

4. Google's Dataset Search Engine

- Google's Dataset Web index helps scientists find and access important datasets from different sources across the web.
- The link is https://toolbox.google.com/datasetsearch.

Popular sources for ML datasets

5. Microsoft Datasets

- Microsoft has launched the **"Microsoft Research Open Data"** repository with the collection of free datasets in various areas such as ***NLP, computer vision, and domain-specific sciences.***
- The link is https://msropendata.com/.

6. Awesome Public Dataset Collection

- Awesome public dataset collection provides high-quality datasets that are arranged in a well-organized manner within a list according to topics such as *Agriculture, Biology, Climate, Complex networks, etc.*
- Most of the datasets are available free, but some may not, so it is better to check the license before downloading the dataset.
- The link is https://github.com/awesomedata/awesome-public-datasets.

Popular sources for ML datasets

7. Government Datasets

- There are different sources to get government-related data.
- Various countries publish government data for public use collected by them from different departments.
- The goal of providing these datasets is to increase the transparency of government work among the people and to use the data in an innovative approach.
- Below are some links to government datasets:
  - Indian Government dataset
  - US Government Dataset
  - Northern Ireland Public Sector Datasets
  - European Union Open Data Portal

Popular sources for ML datasets

8. Computer Vision Datasets

- Visual data provides multiple numbers of great datasets that are specific to computer visions such as ***Image Classification, Video classification, Image Segmentation,*** etc.
- Therefore, if you want to build a project on deep learning or image processing, then you can refer to this source.
- The link is https://www.visualdata.io/.

9. Scikit-learn dataset

- Scikit-learn, a well-known machine learning library in Python, gives a few underlying datasets to practice and trial and error.
- These datasets are open through the sci-kit-learn Programming interface and can be utilized for learning different machine-learning calculations.
- The link is https://scikit-learn.org/stable/datasets/index.html.

Feature selection

- ***Feature Selection*** is one of the important concepts of ML, which highly impacts the performance of the model.
- Feature selection means choosing the most important features (independent variables) from your dataset to improve *model accuracy, reduce overfitting, removing the redundant, irrelevant, or noisy features and speed up training*
- As ML works on the concept of "Garbage In Garbage Out", so we always need to input the most appropriate and relevant dataset to the model in order to get a better result.

Need for Feature Selection

- It helps in avoiding the curse of dimensionality.
- It helps in the simplification of the model so that it can be easily interpreted by the researchers.
- It reduces the training time.
- It reduces overfitting hence enhance the generalization.

Feature Selection: Simple Real-World Example

- Ex: Dataset: Diabetes Prediction
  - Features:
  - Glucose
  - Blood Pressure
  - BMI
  - Age
  - Skin Thickness
  - Insulin

- Using feature selection, we may keep:
  - ✓ Glucose
  - ✓ BMI
  - ✓ Age
- *And drop:*
  - ✗ Skin Thickness
  - ✗ Insulin

  These are removed because they have low importance.

- Ex-2 : Students' Exam Score Prediction
  - Original features:
  - Study hours
  - Sleep hours
  - Height
  - Weight
  - Attendance

❖ After feature selection:
- Study hours
- Attendance

❖ **Removed:**
- Height
- Weight

❖ These do not affect exam score.

Feature Scaling

❖**Feature Scaling** = making all numerical features comparable in range

❖Why Feature Scaling Is Important:

- *Improves model performance:*
  - Many models perform + generalize better when features are scaled.
- *Speeds up model training*
  - Scaling makes optimization easier → faster convergence.
- *Prevents large-value dominance*
  - ✓Ex: Different features have different ranges
    - Income: 3,000 – 50,000
    - Age: 18 – 70
- If not scaled, algorithms give more importance to larger numbers.

Common Methods of Feature Scaling

1. Min–Max Scaling (Normalization)
   - Range becomes **0 to 1**.

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

2. Standardization (Z-score Scaling)
   - Mean becomes **0**, standard deviation b

$$X' = \frac{X - \mu}{\sigma}$$

3. **Robust Scaling**
   - Uses median and IQR → good for **outliers**.

- **Example**
  - Age (years): 20–60
  - Salary (ETB): 3000–40,000
- KNN will treat salary as more important because its scale is bigger.
- After scaling both become comparable, e.g. 0–1 range.

- **Where: X** is the **raw input feature value** you want to scale.

What Do $X_{\min}$ $X_{\max}$,μ ,and σRepresent?

- These are values calculated **from a feature column** (one variable) before scaling

1. $X_{\min}$ →*Minimum value of the feature*
   - The **smallest** value in that column.
   - Example (Age column): [20, 35, 50] $X_{\min}$ =20
2. $X_{\max}$ →***Maximum value of the feature***
   - The **largest** value in that column.
   - Example: (Age column): [20, 35, 50]
     - $X_{\max} = 50$
3. μ**(mu) → Mean (average) of the feature.**
   - The **average** value in the column.
4. σ*(sigma) → Standard deviation of the feature*
   - It tells how spread out the values are from the mean
   - Example (Age column):

$$\mu = \frac{\text{sum of all values}}{\text{number of values}}$$

$$\mu = \frac{20 + 35 + 50}{3} = 35$$

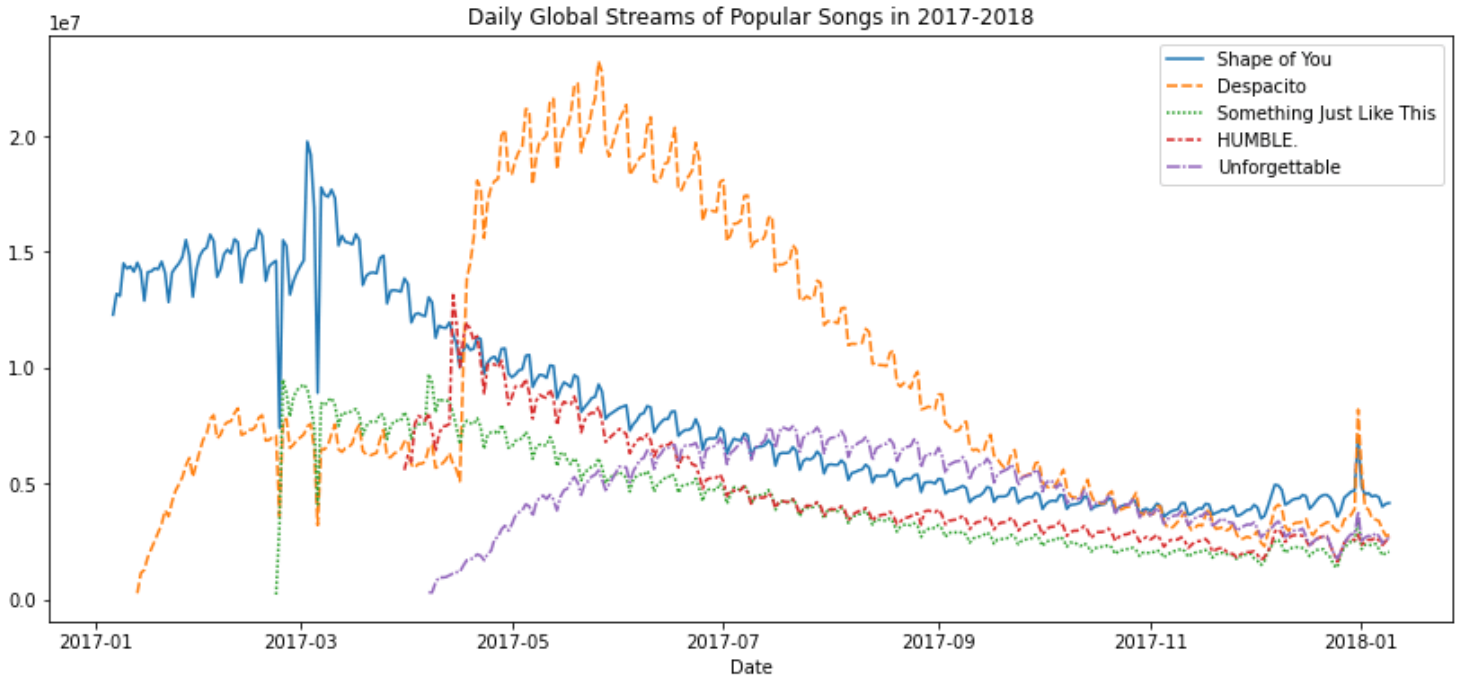
$$\sigma = \sqrt{\frac{(20-35)^2 + (35-35)^2 + (50-35)^2}{3}} = 12.25$$

Data Visualization

- ***Data visualization*** is an essential step in data preparation and analysis as it helps to identify *outliers, trends, and patterns* in the data that may be missed by other forms of analysis.
- With the increasing availability of big data, it has become more important than ever to use data visualization techniques to explore and understand the data.

i. Line Charts:

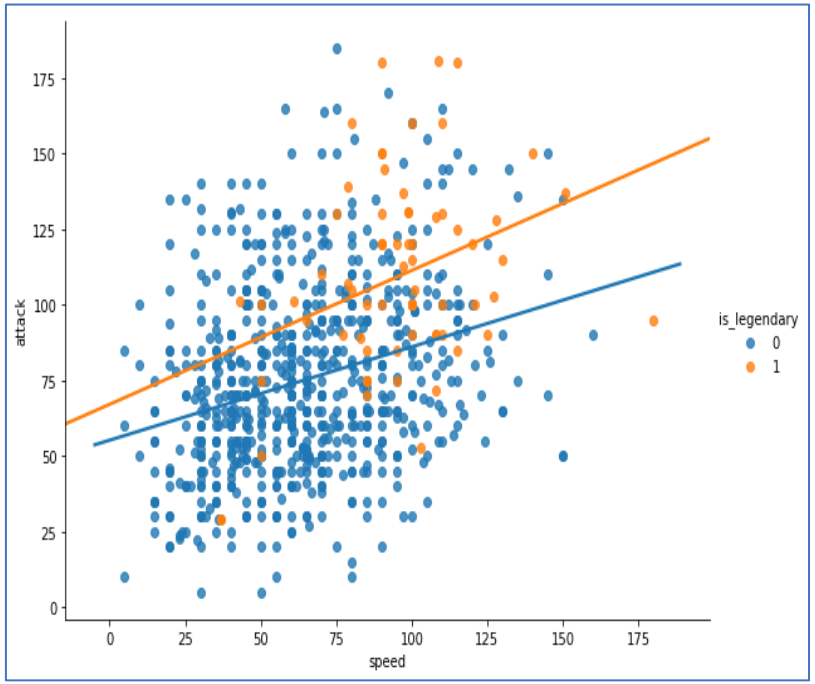
- In a line chart, each data point is represented by a point on the graph, and these points are connected by a line.
- Time-series data is frequently displayed using line charts. For example.



ii. Scatter Plots:

- A quick and efficient method of displaying the relationship between two variables is to use scatter plots.

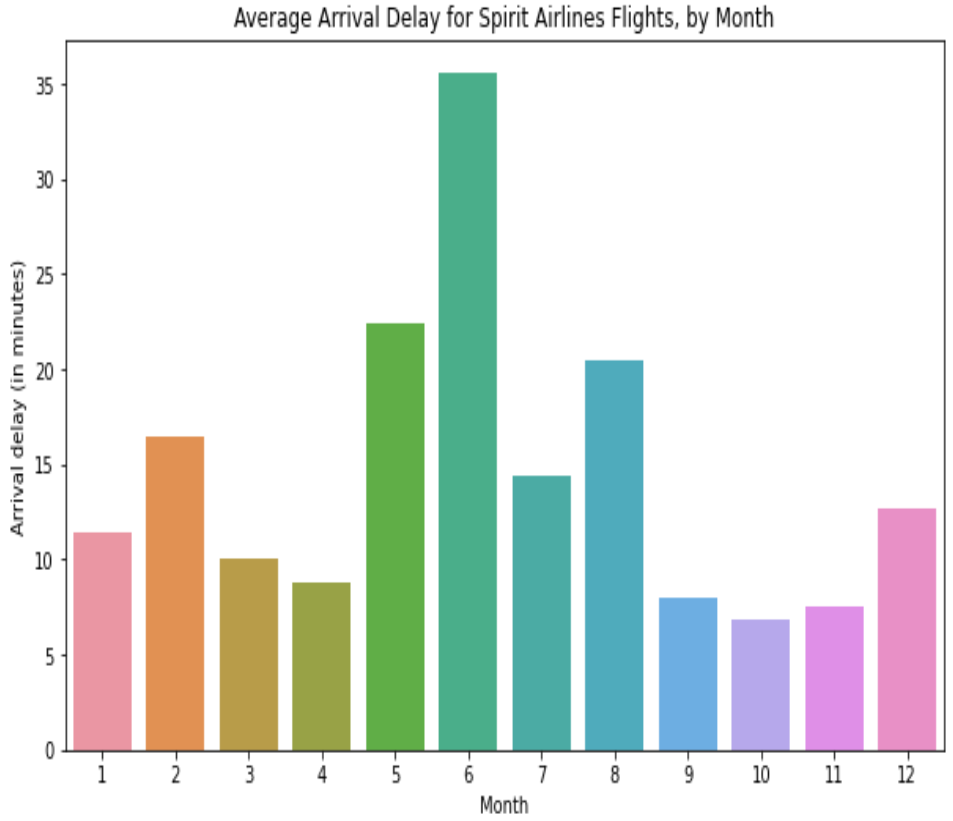
- With one variable plotted on the x-axis and the other variable drawn on the y-axis,
- each data point in a scatter plot is represented by a point on the graph.



- We may use scatter plots to visualize data to find patterns, clusters, and outliers.

iii. Bar Charts

- Bar charts are a common way of displaying categorical data.
- In a bar chart, each category is represented by a bar, with the height of the bar indicating the frequency or proportion of that category in the data.



Note: Other Data visualization approaches: Heat Maps, Tree Maps, Box Plots

Model Evaluation Techniques

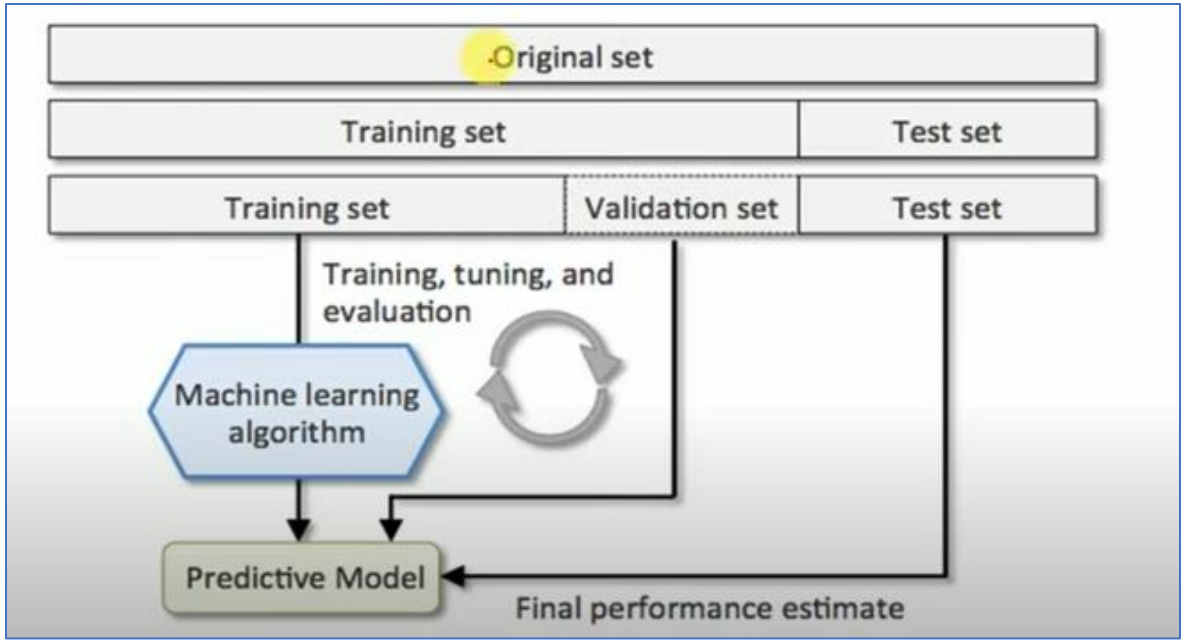
- *In ML, a variety of strategies are employed for model evaluation.*
- *These methods aid in evaluating the functionality and generalizability of trained models.*
- *Typical model evaluation techniques in machine learning include:*

- ***Train-Test Split***
- ***Cross-Validation***
- ***Holdout Validation***

- **Bootstrapping**
- **Shuffled Cross-Validation**

i. Train-Test Split

- In building ML applications,
- datasets are divided into two parts:
  - i. **Training dataset:**
  - ii. **Test Dataset**



- .The training set is used to develop the model, and the ***testing set*** is used to assess it.
- The performance of the model on the *testing set* is used to calculate evaluation measures like ***accuracy and mean squared error.***

ii. Cross-Validation:

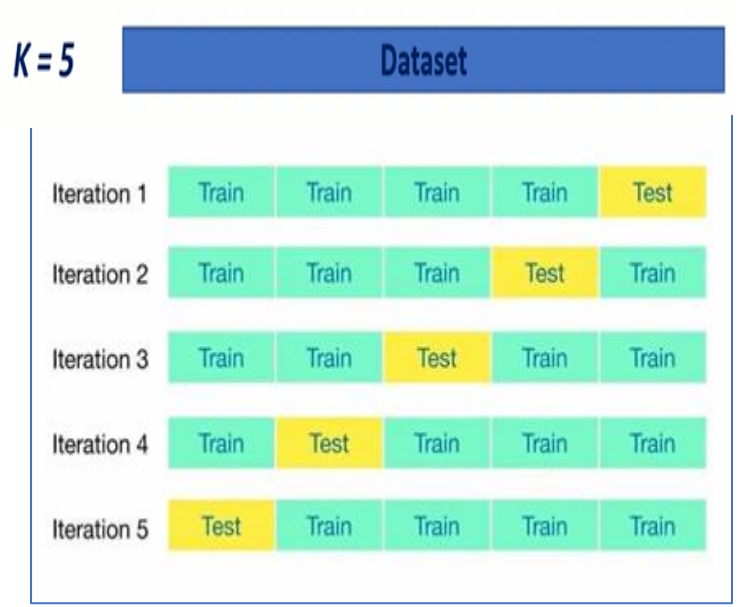
- The idea of cross-validation arises because of the problems with train test model or train test validation model.
- But, the score of our model not depend on the way we picked the train and test.
- **cross-validation** -It entails partitioning the dataset into several folds or subgroups.

- Types of Cross-Validation techniques

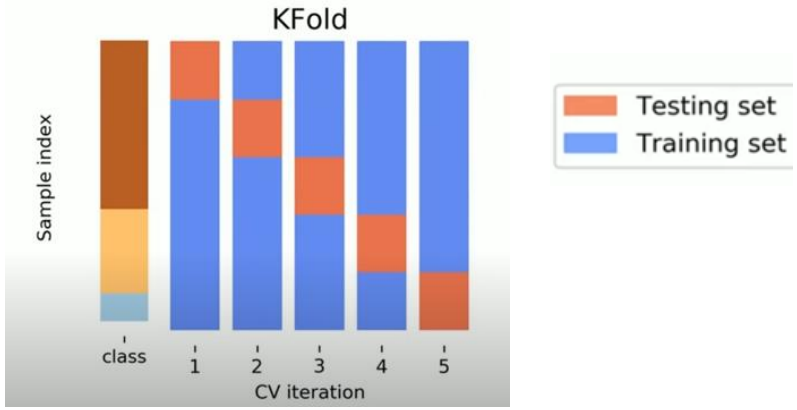
a. K-Fold
b. Stratified K-Folds
c. Leave-One-Out
d. Leave-P-Out

a. K-Fold Cross-Validation

- **K-Fold Cross-Validation** is a model validation technique used in ML to assess how well a model generalizes to unseen data.
- In ***K-Fold Cross Validation,*** we split the dataset into "k" number of folds (subsets).



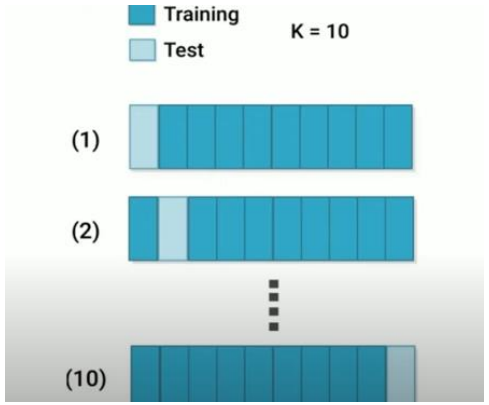
- ***One chunk*** of data is used as test data for evaluation and the remaining part of the data is used for training the model.
- Each time, a different chunk will be used as the test data.



How K-Fold Cross-Validation Works…

1. Divide the Dataset into K Subsets (or Folds):

1. The original dataset is randomly split into **K** equally sized subsets or "folds."
2. Common values for **K** are 5 or 10, but any number can be chosen based on the dataset size.



2.. Train and Validate the Model K Times

- For each of the K iterations, the model is trained on **K-1 folds** (the training set) and tested on the remaining fold (the test set).
- Each fold gets a chance to be the validation set exactly once while being part of the training set **K-1** times.

3. Calculate the Model's Performance:

•After K iterations, you will have **K performance scores**, one for each fold (e.g., accuracy, precision, recall, etc.).

•The final model performance is the **average** of these K scores

Example 1: K-Fold Cross-Validation

- Suppose you have a dataset with 100 data points, and you decide to use **5-fold cross-validation**:

Step 1: Divide the dataset into 5 equal parts, each containing 20 data points.

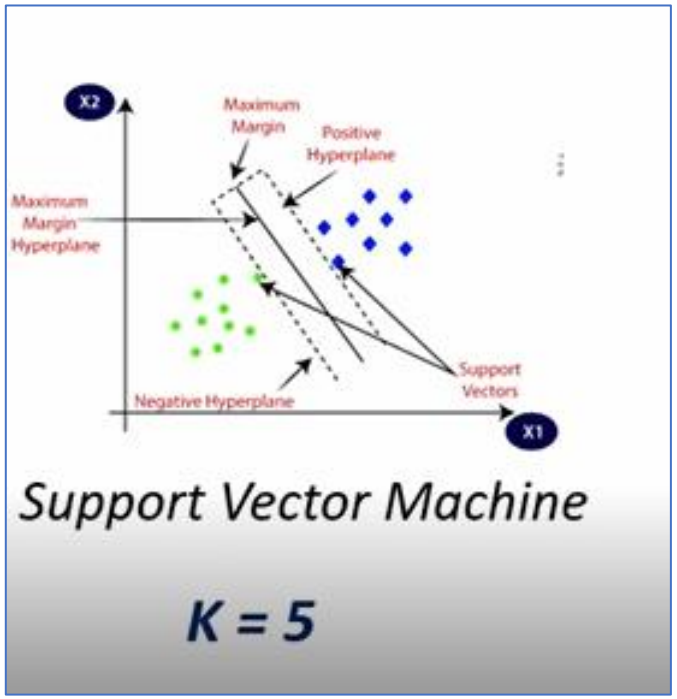
Step 2: For the first iteration, use the first 4 folds (80 data points) for training and the 5th fold (20 data points) for testing.

Step 3: In the second iteration, use folds 1, 2, 3, and 5 for training, and fold 4 for testing.

Step 4: Repeat this process for all 5 folds, each time using a different fold for testing.

Step 5: Calculate the average performance of the model over the 5 iterations.

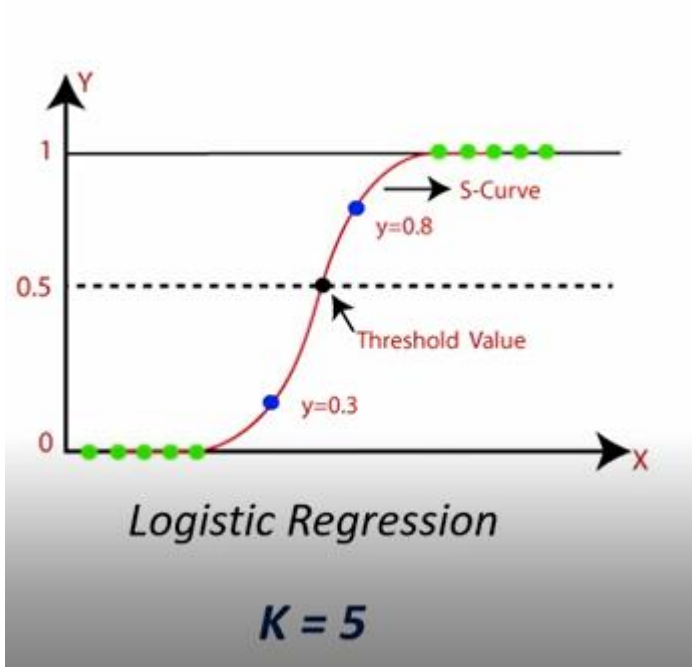
Example 2: **K-Fold Cross-Validation**





$$\textbf{Mean Accuracy} = \frac{88+83+86+81+84}{5} = 84.4\%$$

Example 3: **K-Fold Cross-Validation**



	Dataset					Accuracy
Iteration 1	Train	Train	Train	Train	Test	90%
Iteration 2	Train	Train	Train	Test	Train	88%
Iteration 3	Train	Train	Test	Train	Train	86%
Iteration 4	Train	Test	Train	Train	Train	91%
Iteration 5	Test	Train	Train	Train	Train	85%

$$\textbf{Mean Accuracy} = \frac{90+88+86+91+85}{5} = 88\%$$

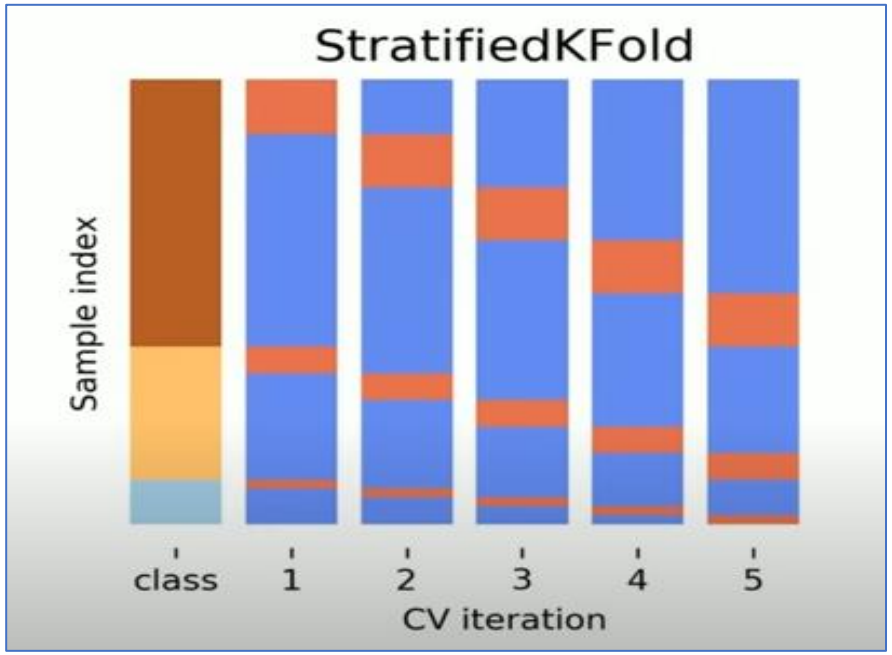
Compare the two Model: K-Fold Cross-Validation

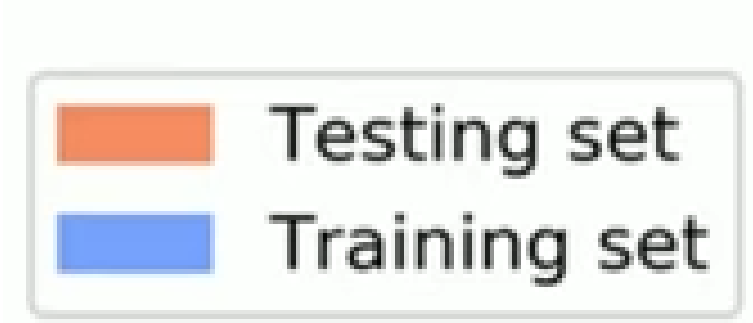
- **Accuracy score for SVM = 84.4%**
- **Accuracy score for logistic Regression = 88%**

- **Advantages of Cross-Validation**
  - *Better alternative for train-test split when the dataset is small.*
  - *Better for multiclass classification problems*
  - *More reliable. Reduced Bias.*
  - *Model Generalization*

b. Stratified K-Fold

- Used when you have ***imbalanced classes***.
- For every iteration, we get equal representation from each class.
- It ensures that **each fold** has approximately the same percentage of samples from each class, the model ***not bias or overfitting*** in this case.





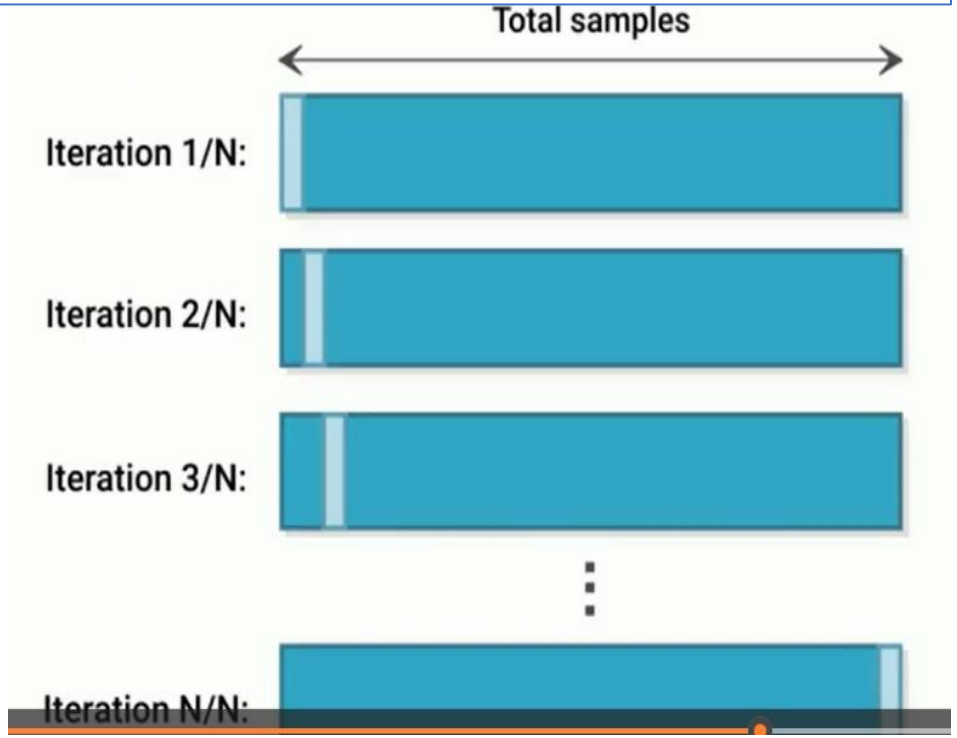
c. Leave-One-Out

- ***Leave-One-Out (LOO)*** is a specific instance of cross-validation in which the model is trained using the remaining data points while each data point is utilized as the evaluation set.

❑*For every iteration we consider 1 examples for testing and n-1 examples for training.*

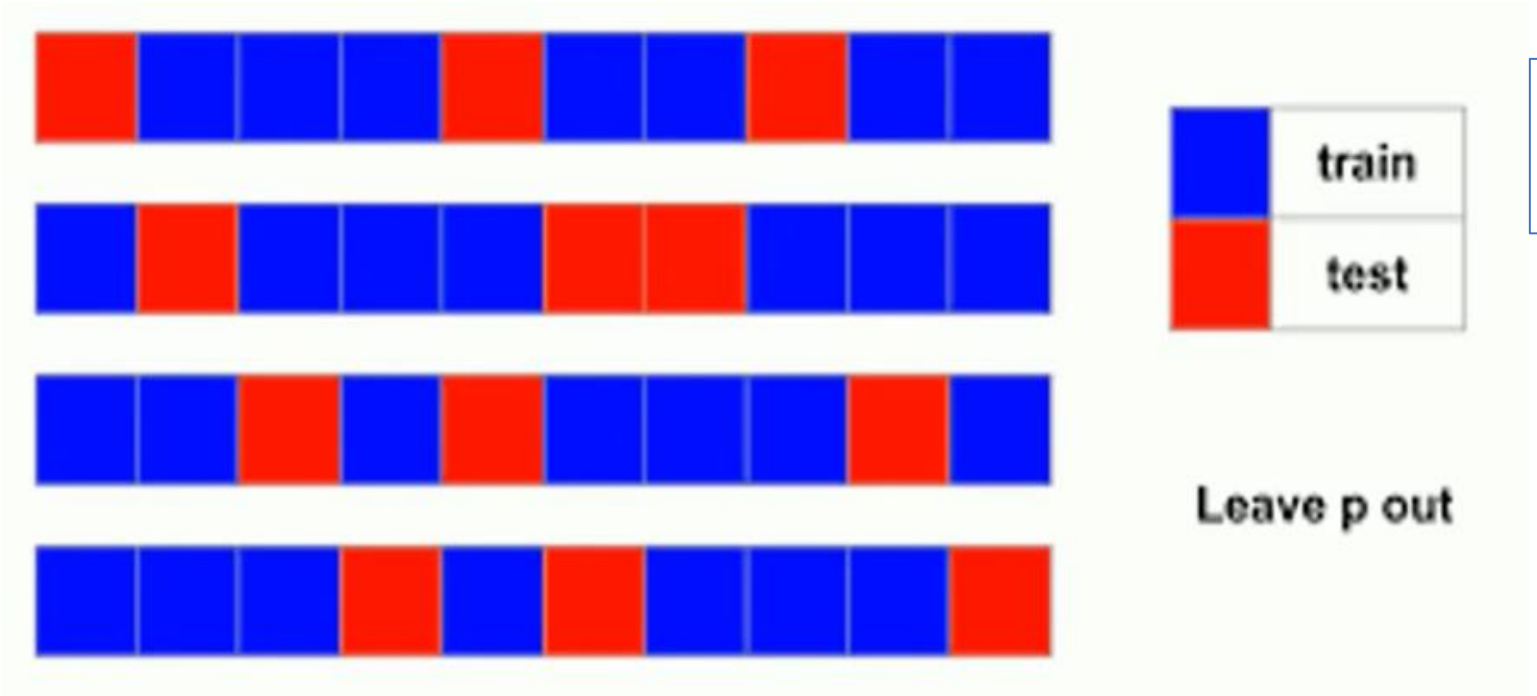
❑*E.g Sample dataset , n = 100, testing =1, training = 100.*

❑*Not used widely because it take a lot of time for training.*



d. Leave-P-Out

- Similar to LOOCV, but instead of leaving out one data point, **P** data points are held out for testing in each iteration.
- We go P example for testing **Iteration**



Example: P= 3(for testing)

iii. Holdout Validation

- Similarly, to the train-test split, ***holdout validation*** entails reserving a unique validating set-in addition to the training & testing sets.
- The training set is used to develop the model, the validation set to fine-tune it, and the testing set to assess it.
- This method aids in picking the top-performing model and fine-tuning the model's hyperparameters.

iv. Bootstrapping

- A ***bootstrapping process*** is a revising approach that entails randomly sampling replacement data from the original dataset in order to produce numerous datasets.
- A ***model*** is trained and evaluated using each bootstrap sample, and the performance of the model is estimated using an aggregate of the evaluation results.
- A measure of the model's *reliability* & ***robustness*** can be obtained by bootstrapping.

v. Shuffled Cross-Validation

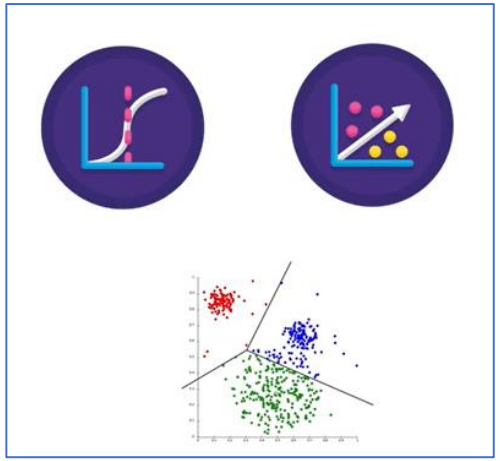
- This method extends k-fold cross-validation by randomly shuffling the data before the cross-validation step.
- It helps to minimize any potential ***bias*** induced by the ordering of the data by ensuring that the data distribution is more evenly represented across the folds.

Model selection

- ***Model Selection*** is the process of choosing the best suited model for a particular problem.
- Selecting a model depends on various factors such as the dataset, task, nature of the model, etc

Two factors to be considered:

1. Logical Reason to select a model
2. Comparing the performance of the models



Model Selection

Models can be selected based on:

1. **Type of Data available:**
    a. Images and videos-CNN
    b. Text data or Speech data-RNN
    c. Numerical data-SVM, Logistic Regression, Decision Trees, etc.
2. Based on the task we need to carry out:
    b. Classification tasks-SVM, Logistic Regression, Decision Trees, etc.
    c. Regression tasks-Linear regression, Polynomial regression etc,
    d. Clustring tasks- k-means clustering, hierarchical clustering

Over-fitting and Under-fitting

- **Overfitting** and underfitting occur while training our machine learning or deep learning models, both are the main problems that occur in **ML or DL.**
- The main goal of each ML model is ***to generalize well***.
- **Generalization** defines the ability of an ***ML model*** to provide a suitable output by adapting the given set of unknown inputs.
- It means after providing training on the dataset, it can produce reliable and accurate output.
- Hence, the two terms that need to be checked for the performance of the model and whether the model is ***generalizing well or not.***

Over-fitting and Under-fitting

❑*Some basic terms that will help to understand this topic well:*

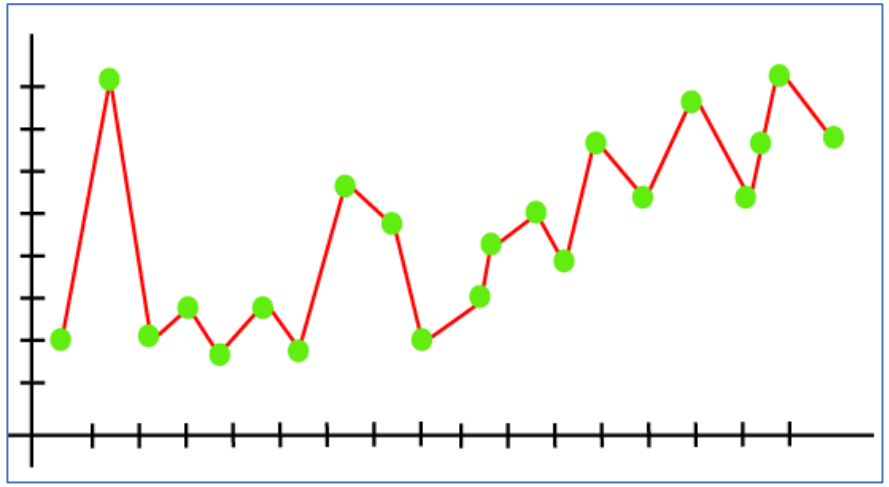
- ❖**Signal:** It refers to the true underlying pattern of the data that helps the machine learning model to learn from the data.
- ❖**Noise:-** is unnecessary and irrelevant data that reduces the performance of the model.
- ❖**Bias:-** is a prediction error that is introduced in the model due to oversimplifying the ML algorithms.
- ❖ Or it is the difference between the ***predicted values*** and the ***actual values***.
- ❖**Variance:** If the ML model performs well with the ***training dataset***, but does not perform well with the ***test dataset***, then ***variance occurs***.

Over-fitting

- **Overfitting** occurs when our ML model ***tries to cover all the data points*** .
- i.e. our training has focused on the particular training set so much that it has missed the point entirely.
- Because of this, the model starts caching ***noise*** and ***inaccurate values*** present in the dataset, and all these factors reduce the efficiency and accuracy of the model.
- The **overfitted** model has ***low bias*** and ***high variance*.**
- The chances of occurrence of overfitting increase as much as we provide training to our model.
- It means the more we train our model, the more chances of occurring the overfitted model.

Over-fitting : Example

- Consider the below graph of the linear regression output:



- As we can see from the above graph, the model tries to cover all the data points present in the scatter plot.
- It may look efficient, but in reality, it is not so. Because the goal of the regression model to find the best fit line, but here we have not got any best fit, so, it will generate the prediction errors.

How to Avoid the overfitting in Model

- Both overfitting and underfitting cause the degraded performance of the ML model.
- But the main cause is ***overfitting***, so there are some ways by which we can reduce the occurrence of overfitting in our model.

- ***Cross-Validation***
- ***Training with more data***
- ***Removing features***

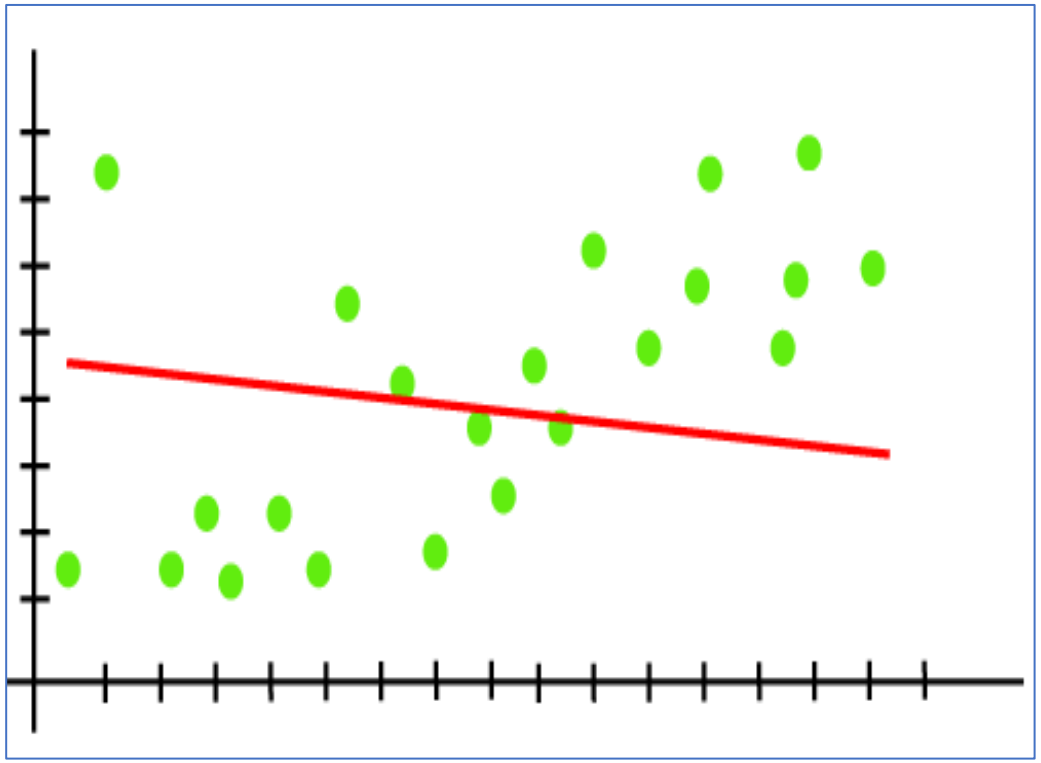
- ***Early stopping the training***
- ***Regularization***
- ***Ensembling***

Under-fitting

- **Underfitting** occurs when our ML model is not able to capture the *underlying trend of the data.*
- It doesn't know what to do with the task we've given it and, therefore, provides an answer that is far from correct.
- **In the case of underfitting,** the model is not able to learn enough from the training data, and hence it *reduces the accuracy and produces unreliable predictions.*
- An underfitted model has *high bias and low variance*.

Under-fitting : Example

- As we can see from the diagram, the model is unable to capture the data points present in the plot.



- How to avoid underfitting:
  - *By increasing the training time of the model.*
  - *By increasing the number of features.*

Goodness of Fit

- The **"Goodness of fit"** term is taken from the statistics, and the goal of the ML models to achieve the *goodness of fit.*
- In statistics modeling, *it defines how closely the result or predicted values match the true values of the dataset.*
- The model with a good fit is between the ***underfitted and overfitted model***, and ideally, it makes predictions with 0 errors, but in practice, it is difficult to achieve it.
- As when we *train our model* for a time, the errors in the training data go down, and the same happens with test data.
- But if we train the model for a long duration, then the performance of the model may decrease due to the overfitting, as the model also learn the noise present in the dataset.
- The errors in the test dataset start increasing, *so the point, just before the raising of errors, is the good point, and we can stop here for achieving a good model.*

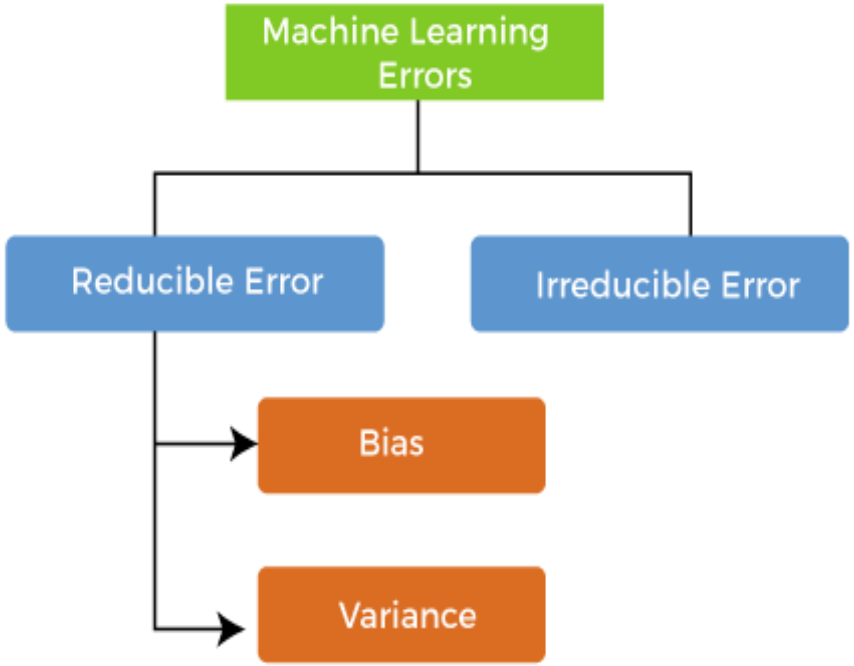
Bias and Variance

- If the ML model is not accurate, it can make prediction errors, and these prediction errors are usually known as **Bias and Variance.**
- In ML , these errors will always be present as there is a difference between the ***model predictions and actual predictions.***
- The main aim of ***ML/data science analysts*** is to reduce these errors to get more accurate results.
- Let's first understand **what Errors** in Machine learning.
- **Errors In ML**
  - In ML, an ***error*** is a measure of how accurately an algorithm can make predictions for the previously unknown dataset.
  - On the basis of these errors, the ML model is selected that can perform best on the particular dataset.
  - There are mainly two types of errors in machine learning, which are:

Types of errors in ML

i. Reducible errors: These errors can be reduced to improve the model accuracy.

- Such errors can further be classified into ***bias and Variance***.



ii. Irreducible errors: These errors will always be present in the model regardless of which algorithm has been used.

- The cause of these errors is unknown variables whose value can't be reduced.

What is Bias?

- In general, the ML model analyses the data, finds patterns in it, and makes predictions.
- While training, the model learns these patterns in the dataset and applies them to test data for prediction.
  - ***While making predictions, a difference occurs between prediction values made by the model and actual values/expected values, and this difference is known as bias errors or Errors due to bias.***
- **A model has either:**
  - **Low Bias:** A low-bias model will make fewer assumptions about the form of the target function.
  - **High Bias:** A model with a high bias makes more assumptions, and the model becomes unable to capture the important features of our dataset.
  - **A high bias model also cannot perform well on new data.**

What is a Variance Error?

- ***Variance*** *tells that how much a random variable is different from its expected value.*
- Variance errors are either of low variance or high variance.
  - **Low variance** means there is a small variation in the prediction of the target function with changes in the training data set.
  - High variance shows a large variation in the prediction of the target function with changes in the training dataset.
- A model that shows high variance learns a lot and perform well with the training dataset, and does not generalize well with the unseen dataset.
- As a result, such a model gives good results with the training dataset but shows high error rates on the test dataset.

What is a Variance Error?

- A model with **high variance** has the following problems:
  - *A high variance model leads to overfitting.*
  - *Increase model complexities.*
  - *Usually, nonlinear algorithms have a lot of flexibility to fit the model, and have high variance.*
- Ways to Reduce High Variance:
  - *Reduce the input features or number of parameters as a model is overfitted.*
  - *Do not use a much complex model.*
  - *Increase the training data.*
  - *Increase the Regularization term.*